

The Rounded-Polygon Packing Model

A Complete Derivation

Mathematical foundations of pyCudaPolygon

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Abstract

This is a self-contained derivation of every algorithm used to relax a dense packing of soft, equilateral, inward-rounded polygons under periodic boundary conditions. Nothing is assumed beyond multivariable calculus and linear algebra. The geometry is carried entirely in index notation with the two-dimensional Levi-Civita symbol, which reduces all signed areas, cross products and rotations to two contraction identities. We build the model in the order the code uses it: polygon geometry, the inscribed rounding and its arc geometry, the overlap area as a soft potential via Green's theorem, the analytic force as the gradient of that area, SHAKE constraint projection, force projection onto the constraint manifold, FIRE relaxation, random shape generation by SLSQP, and the bidisperse packing set-up. Every intermediate step is shown.

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1 Notation and the tensor toolkit

1.1 Index conventions

Latin indices $i, j, k, l, m, p, q \in \{1, 2\}$ label Cartesian components in the plane. The Einstein summation convention applies to repeated component indices and to nothing else; in particular vertex labels (written as subscripts $a, b, \dots$ or k) are never summed implicitly. The plane carries the Euclidean metric δ_{ij} , so raising and lowering indices is the identity and is used only for visual balance. A point or vector $\mathbf{u}$ has components u^i or u_i interchangeably; $|\mathbf{u}| = \sqrt{\delta_{ij}u^i u^j}$.

1.2 The two-dimensional Levi-Civita symbol

Define the totally antisymmetric symbol

$$\epsilon_{ij} = -\epsilon_{ji}, \quad \epsilon_{12} = +1, \quad \epsilon_{11} = \epsilon_{22} = 0, \quad (1.1)$$

and ϵ^{ij} numerically equal to ϵ_{ij} . Its entire utility rests on two contraction identities.

Lemma 1 (Contraction identities).

$$\boxed{\epsilon_{ij} \epsilon_{lm} = \delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}, \quad \epsilon_{ij} \epsilon_{lj} = \delta_{il}.} \quad (1.2)$$

Proof. Both sides of the first identity are antisymmetric in $i \leftrightarrow j$ and in $l \leftrightarrow m$, and symmetric under the pair swap $(ij) \leftrightarrow (lm)$. In two dimensions an antisymmetric pair must be $(1, 2)$ or $(2, 1)$, so it suffices to check $(ij) = (12)$. Then the left side is $\epsilon_{12} \epsilon_{lm} = \epsilon_{lm}$, nonzero only for $(lm) = (12)$ giving $+1$ and $(lm) = (21)$ giving -1 . The right side is $\delta_{1l} \delta_{2m} - \delta_{1m} \delta_{2l}$, which is $+1$ at $(lm) = (12)$, -1 at $(lm) = (21)$, and 0 otherwise. They agree. Contracting $m = j$ in the first identity and summing, $\epsilon_{ij} \epsilon_{lj} = \delta_{il} \delta_{jj} - \delta_{ij} \delta_{jl} = 2\delta_{il} - \delta_{il} = \delta_{il}$, using $\delta_{jj} = 2$ in the plane. $\square$

1.3 Products, rotations, and their calculus

For vectors $\mathbf{u}, \mathbf{v}$:

$$\text{dot product:} \quad \mathbf{u} \cdot \mathbf{v} = \delta_{ij} u^i v^j, \quad (1.3)$$

$$\text{scalar cross product:} \quad \mathbf{u} \times \mathbf{v} := \epsilon_{ij} u^i v^j = u^1 v^2 - u^2 v^1, \quad (1.4)$$

$$+90^\circ \text{ rotation:} \quad (\mathbf{u}^\perp)^i := -\epsilon^{ij} u_j, \quad \text{so} \quad \mathbf{u}^\perp = (-u^2, u^1). \quad (1.5)$$

The rotation $\mathbf{u} \mapsto \mathbf{u}^\perp$ is the proper (counter-clockwise) quarter turn; it is the inward normal to a directed edge and matches the convention $\hat{\mathbf{n}} = (-\hat{u}_y, \hat{u}_x)$. The following identities follow directly from Lemma 1 and are used repeatedly:

$$\mathbf{u}^\perp \cdot \mathbf{u} = 0, \quad |\mathbf{u}^\perp| = |\mathbf{u}|, \quad \mathbf{u} \times \mathbf{v} = \mathbf{u}^\perp \cdot \mathbf{v} = -\mathbf{u} \cdot \mathbf{v}^\perp, \quad (\mathbf{u}^\perp)^\perp = -\mathbf{u}. \quad (1.6)$$

We will differentiate norms, unit vectors and the two-argument arctangent so often that they are isolated here once and for all.

Lemma 2 (Calculus of vector normalisation). For $\mathbf{u} \neq \mathbf{0}$ with $\hat{\mathbf{u}} = \mathbf{u}/|\mathbf{u}|$,

$$\boxed{\frac{\partial |\mathbf{u}|}{\partial u^k} = \hat{u}_k, \quad \frac{\partial \hat{u}^j}{\partial u^k} = \frac{\delta^{jk} - \hat{u}^j \hat{u}^k}{|\mathbf{u}|} := \frac{P^{jk}[\hat{\mathbf{u}}]}{|\mathbf{u}|},} \quad (1.7)$$

where $P^{jk}[\hat{\mathbf{u}}] = \delta^{jk} - \hat{u}^j \hat{u}^k$ is the orthogonal projector off $\hat{\mathbf{u}}$ ($P \hat{\mathbf{u}} = 0$, $P^2 = P$, $P^\top = P$).

Proof. $|\mathbf{u}|^2 = \delta_{lm} u^l u^m$, so $2|\mathbf{u}| \partial|\mathbf{u}|/\partial u^k = 2u_k$, giving $\partial|\mathbf{u}|/\partial u^k = u_k/|\mathbf{u}| = \hat{u}_k$. Then

$$\frac{\partial \hat{u}^j}{\partial u^k} = \frac{\partial}{\partial u^k} (u^j |\mathbf{u}|^{-1}) = \frac{\delta^{jk}}{|\mathbf{u}|} + u^j (-|\mathbf{u}|^{-2}) \hat{u}_k = \frac{\delta^{jk}}{|\mathbf{u}|} - \frac{\hat{u}^j \hat{u}^k}{|\mathbf{u}|} = \frac{\delta^{jk} - \hat{u}^j \hat{u}^k}{|\mathbf{u}|}.$$

$P\hat{\mathbf{u}} = 0$ and $P^2 = P$ are immediate from $\hat{u}^j \hat{u}^j = 1$. $\square$

Lemma 3 (Derivative of the planar argument). If c and d are differentiable functions of a parameter and $\psi = \text{atan2}(c, d)$ (the angle of (d, c) , branch-free), then

$$\boxed{d\psi = \frac{d \, dc - c \, dd}{c^2 + d^2}.} \quad (1.8)$$

In particular, if $(d, c) = (\cos \psi, \sin \psi)$ lies on the unit circle then $c^2 + d^2 = 1$ and $d\psi = d \, dc - c \, dd$.

Proof. Away from the origin ψ satisfies $\tan \psi = c/d$ on each branch and is C^∞ . Differentiate $c = \rho \sin \psi$, $d = \rho \cos \psi$ with $\rho = \sqrt{c^2 + d^2}$: then $d \, dc - c \, dd = \rho \cos \psi (d\rho \sin \psi + \rho \cos \psi \, d\psi) - \rho \sin \psi (d\rho \cos \psi - \rho \sin \psi \, d\psi) = \rho^2 d\psi$. Divide by $\rho^2 = c^2 + d^2$. $\square$

Lemma 4 (Half-angle identity). For $\phi \in (-\pi, \pi)$, $\tan \frac{\phi}{2} = \frac{\sin \phi}{1 + \cos \phi} = \frac{1 - \cos \phi}{\sin \phi}$.

Proof. $\sin \phi = 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2}$ and $1 + \cos \phi = 2 \cos^2 \frac{\phi}{2}$, so the first ratio is $\sin \frac{\phi}{2} / \cos \frac{\phi}{2} = \tan \frac{\phi}{2}$. Likewise $1 - \cos \phi = 2 \sin^2 \frac{\phi}{2}$ gives the second. $\square$

1.4 Periodic box

All vertices live in the torus $[0, 1)^2$. Two distinct maps act componentwise: the *minimum-image* map for separations and the *fold* for absolute positions,

$$\text{mic}(x) = x - \lfloor x + \tfrac{1}{2} \rfloor \in [-\tfrac{1}{2}, \tfrac{1}{2}), \quad \text{fold}(x) = x - \lfloor x \rfloor \in [0, 1). \quad (1.9)$$

A displacement between two points is always reduced with mic so that the shorter of the two periodic images is taken; an integrated position is returned to the box with fold.

Every signed area, edge cross product and 90° rotation below is one operation: contraction with ϵ_{ij} . Lemmas 1–4 are the only analytic facts needed; the rest is bookkeeping. Keeping $P^{jk} = \delta^{jk} - \hat{u}^j \hat{u}^k$ in mind explains why every force that flows through a unit edge vector comes out automatically perpendicular to that edge.

2 Polygon geometry

A backbone polygon P has N vertices $\mathbf{a}_0, \dots, \mathbf{a}_{N-1} \in [0, 1)^2$, indexed mod N and traversed counter-clockwise. Edge E_k runs from $\mathbf{a}_k$ to $\mathbf{a}_{k+1}$. Within an interaction all coordinates are placed in one minimum-image frame anchored at $\mathbf{a}_0$, after which ordinary (non-periodic) calculus applies. The incoming and outgoing edge vectors at vertex k and their unit versions are

$$u_k^i = \text{mic}(a_k^i - a_{k-1}^i), \quad v_k^i = \text{mic}(a_{k+1}^i - a_k^i), \quad \hat{\mathbf{u}}_k = \frac{\mathbf{u}_k}{|\mathbf{u}_k|}, \quad \hat{\mathbf{v}}_k = \frac{\mathbf{v}_k}{|\mathbf{v}_k|}. \quad (2.1)$$

2.1 Signed area from Green's theorem

Theorem 1 (Green, shoelace form). For a piecewise- C^1 region Ω with positively (CCW) oriented boundary,

$$\text{Area}(\Omega) = \frac{1}{2} \oint_{\partial\Omega} \epsilon_{ij} x^i dx^j = \frac{1}{2} \oint_{\partial\Omega} (x dy - y dx). \quad (2.2)$$

Proof. Green's theorem states $\oint_{\partial\Omega} (L dx + M dy) = \iint_{\Omega} (\partial_x M - \partial_y L) dA$. Take $L = -\frac{1}{2}y$, $M = \frac{1}{2}x$; then $\partial_x M - \partial_y L = \frac{1}{2} + \frac{1}{2} = 1$, so the line integral equals $\iint_{\Omega} 1 dA = \text{Area}(\Omega)$. Writing the integrand as $\frac{1}{2}(x dy - y dx) = \frac{1}{2}\epsilon_{ij}x^i dx^j$ gives the stated form. $\square$

Apply Theorem 1 to the polygon. The boundary is the union of edges E_k , on which $\mathbf{x}(s) = (1-s)\mathbf{a}_k + s\mathbf{a}_{k+1}$, $s \in [0, 1]$. A single edge contributes (this computation recurs verbatim later, so we do it once and in full):

$$\begin{aligned} \int_{E_k} (x dy - y dx) &= \int_0^1 \left[((1-s)a_k^x + s a_{k+1}^x)(a_{k+1}^y - a_k^y) - ((1-s)a_k^y + s a_{k+1}^y)(a_{k+1}^x - a_k^x) \right] ds \\ &= \frac{1}{2}(a_k^x + a_{k+1}^x)(a_{k+1}^y - a_k^y) - \frac{1}{2}(a_k^y + a_{k+1}^y)(a_{k+1}^x - a_k^x) \\ &= a_k^x a_{k+1}^y - a_k^y a_{k+1}^x = \epsilon_{ij} a_k^i a_{k+1}^j, \end{aligned} \quad (2.3)$$

where the diagonal pieces $a_k^x a_k^y$ and $a_{k+1}^x a_{k+1}^y$ cancelled by the antisymmetry of ϵ_{ij} . Summing over edges and halving:

$$A_{\text{bb}} = \frac{1}{2} \sum_{k=0}^{N-1} \epsilon_{ij} a_k^i a_{k+1}^j, \quad (2.4)$$

positive for CCW orientation. This is the shoelace formula.

2.2 Gradient of the area

Each vertex $\mathbf{a}_k$ appears in exactly two terms of (2.4): as the first factor of the k -th term and the second factor of the $(k-1)$ -th. Hence

$$g_k^i := \frac{\partial A_{\text{bb}}}{\partial a_k^i} = \frac{1}{2} (\epsilon_{ij} a_{k+1}^j + \epsilon_{ji} a_{k-1}^j) = \left[\frac{1}{2} \epsilon_{ij} (a_{k+1}^j - a_{k-1}^j) \right], \quad (2.5)$$

using $\epsilon_{ji} = -\epsilon_{ij}$. Componentwise this is $\partial A / \partial a_k^x = \frac{1}{2}(a_{k+1}^y - a_{k-1}^y)$ and $\partial A / \partial a_k^y = \frac{1}{2}(a_{k-1}^x - a_{k+1}^x)$. The vector $\mathbf{g}_k = \frac{1}{2}(\mathbf{a}_{k-1} - \mathbf{a}_{k+1})^\perp$ (equivalently $-\frac{1}{2}(\mathbf{a}_{k+1} - \mathbf{a}_{k-1})^\perp$, using the convention $\mathbf{u}^\perp = (-u^2, u^1)$ of §1.3) is the area-constraint gradient used by SHAKE.

2.3 Turning angle and shape index

At vertex k define the two rotation invariants of the incoming/outgoing frame,

$$c_k := \hat{\mathbf{u}}_k \times \hat{\mathbf{v}}_k = \epsilon_{ij} \hat{u}_k^i \hat{v}_k^j, \quad d_k := \hat{\mathbf{u}}_k \cdot \hat{\mathbf{v}}_k = \delta_{ij} \hat{u}_k^i \hat{v}_k^j, \quad c_k^2 + d_k^2 = 1, \quad (2.6)$$

so that $c_k = \sin \phi_k$, $d_k = \cos \phi_k$, where the exterior (turning) angle is

$$\phi_k = \text{atan2}(c_k, d_k). \quad (2.7)$$

Theorem 2 (Theorem of turning tangents). For a simple closed polygon traversed once counter-clockwise, $\sum_{k=0}^{N-1} \phi_k = 2\pi$.

Proof sketch. Smooth each corner by a small circular fillet of vanishing radius. The unit tangent of the resulting C^1 closed curve is a continuous map $S^1 \rightarrow S^1$; its degree (the winding number of the tangent) is the total turning $\frac{1}{2\pi} \oint d\theta$. For a simple curve this degree is the winding number of the curve about an interior point, namely $+1$ for CCW orientation (Hopf's *Umlaufsatz*). Each fillet contributes its turning ϕ_k and the straight segments contribute nothing, so $\sum_k \phi_k = 2\pi$. $\square$

For a convex CCW polygon every $\phi_k > 0$. SHAKE enforces the equilateral constraint $|\mathbf{v}_k| = L_k$ (equal edge lengths, *not* equal angles); with all $L_k = L$ the polygon is equilateral. The *shape index* (reduced perimeter) is

$$\kappa = \frac{P}{\sqrt{A_{\text{bb}}}}, \quad P = \sum_k L_k, \quad (2.8)$$

which is $2\sqrt{\pi} \approx 3.545$ for a circle, 4 for a square, and larger for elongated shapes; it is dimensionless and scale-invariant.

3 The inscribed rounded corner

At every vertex the sharp corner is replaced by a circular arc of radius $\delta > 0$, tangent to both incident edges and curving toward the interior. The rounded boundary $\partial\tilde{P}$ alternates arcs K_k and straight sections E_k . This is an *inscribed* rounding — it removes material at each corner — and is distinct from the outward Minkowski sum $P \oplus D(\delta)$.

3.1 The corner frame

Fix a vertex and drop its index. Let $\hat{\mathbf{n}} = \hat{\mathbf{u}}^\perp$ be the inward unit normal to the incoming edge. Then $\{\hat{\mathbf{u}}, \hat{\mathbf{n}}\}$ is a positively oriented orthonormal frame, and the outgoing unit direction resolves as

$$\hat{\mathbf{v}} = (\hat{\mathbf{v}} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}} + (\hat{\mathbf{v}} \cdot \hat{\mathbf{n}}) \hat{\mathbf{n}} = d \hat{\mathbf{u}} + c \hat{\mathbf{n}}, \quad (3.1)$$

because $\hat{\mathbf{v}} \cdot \hat{\mathbf{u}} = d$ and $\hat{\mathbf{v}} \cdot \hat{\mathbf{n}} = \hat{\mathbf{v}} \cdot \hat{\mathbf{u}}^\perp = -\hat{\mathbf{u}} \times \hat{\mathbf{v}} \cdot (-1) = \hat{\mathbf{u}} \times \hat{\mathbf{v}} = c$ (using $\mathbf{u}^\perp \cdot \mathbf{v} = \mathbf{u} \times \mathbf{v}$, here with $\mathbf{u} = \hat{\mathbf{u}}$). Consistency: $|\hat{\mathbf{v}}|^2 = d^2 + c^2 = 1$. $\checkmark$

3.2 Tangent set-back and the arc centre

Write the unknown centre as

$$\mathbf{z} = \mathbf{a} + \alpha \hat{\mathbf{u}} + \beta \hat{\mathbf{n}}, \quad (3.2)$$

and let the tangent points be $\mathbf{a}^- = \mathbf{a} - \ell \hat{\mathbf{u}}$ on the incoming edge and $\mathbf{a}^+ = \mathbf{a} + \ell \hat{\mathbf{v}}$ on the outgoing edge, with common set-back $\ell > 0$.

Tangency on the incoming edge. The circle of radius δ is tangent to the incoming edge at $\mathbf{a}^-$ iff the radius $\mathbf{z} - \mathbf{a}^-$ is perpendicular to the edge direction $\hat{\mathbf{u}}$, i.e. has no $\hat{\mathbf{u}}$ -component:

$$\mathbf{z} - \mathbf{a}^- = (\alpha + \ell) \hat{\mathbf{u}} + \beta \hat{\mathbf{n}} \implies (\mathbf{z} - \mathbf{a}^-) \cdot \hat{\mathbf{u}} = \alpha + \ell = 0 \implies \boxed{\alpha = -\ell}. \quad (3.3)$$

Tangency on the outgoing edge. Likewise $\mathbf{z} - \mathbf{a}^+$ must be perpendicular to $\hat{\mathbf{v}}$. Using (3.1) and $\alpha = -\ell$,

$$\begin{aligned} \mathbf{z} - \mathbf{a}^+ &= \alpha \hat{\mathbf{u}} + \beta \hat{\mathbf{n}} - \ell \hat{\mathbf{v}}, \\ (\mathbf{z} - \mathbf{a}^+) \cdot \hat{\mathbf{v}} &= \alpha (\hat{\mathbf{u}} \cdot \hat{\mathbf{v}}) + \beta (\hat{\mathbf{n}} \cdot \hat{\mathbf{v}}) - \ell (\hat{\mathbf{v}} \cdot \hat{\mathbf{v}}) = \alpha d + \beta c - \ell = -\ell d + \beta c - \ell = 0, \\ \implies \beta &= \frac{\ell(1+d)}{c}. \end{aligned} \quad (3.4)$$

Radius condition. Because (3.3) annihilates the $\hat{\mathbf{u}}$ -component of $\mathbf{z} - \mathbf{a}^-$, the radius vector is purely normal: $\mathbf{z} - \mathbf{a}^- = \beta \hat{\mathbf{n}}$, whence

$$\delta^2 = |\mathbf{z} - \mathbf{a}^-|^2 = \beta^2 \implies \beta = \delta \quad (\beta > 0 \text{ for a convex CCW corner, } c > 0). \quad (3.5)$$

No quadratic in ℓ arises: the set-back is fixed directly by combining (3.4) and (3.5).

With $c = \sin \phi$, $d = \cos \phi$ and Lemma 4,

$$\ell = \frac{\delta c}{1+d} = \delta \frac{\sin \phi}{1+\cos \phi} = \delta \tan \frac{\phi}{2}, \quad \mathbf{z} = \mathbf{a} - \ell \hat{\mathbf{u}} + \delta \hat{\mathbf{n}}. \quad (3.6)$$

Equivalent centre formula. The form used in the kernel is $\mathbf{z} = \mathbf{a} + \frac{\delta(\hat{\mathbf{v}} - \hat{\mathbf{u}})}{|c|}$. To verify it equals (3.6), resolve $\hat{\mathbf{v}} - \hat{\mathbf{u}}$ in the corner frame using (3.1): $\hat{\mathbf{v}} - \hat{\mathbf{u}} = (d-1)\hat{\mathbf{u}} + c\hat{\mathbf{n}}$, so for $c > 0$

$$\frac{\delta(\hat{\mathbf{v}} - \hat{\mathbf{u}})}{|c|} = \frac{\delta(d-1)}{c} \hat{\mathbf{u}} + \delta \hat{\mathbf{n}}.$$

Now $\frac{d-1}{c} = -\frac{1-d}{c} = -\frac{(1-d)(1+d)}{c(1+d)} = -\frac{1-d^2}{c(1+d)} = -\frac{c^2}{c(1+d)} = -\frac{c}{1+d} = -\tan \frac{\phi}{2}$, using $1-d^2 = c^2$. Hence the $\hat{\mathbf{u}}$ -coefficient is $-\delta \tan(\phi/2) = -\ell$, and the formula reproduces $\mathbf{z} - \mathbf{a} = -\ell \hat{\mathbf{u}} + \delta \hat{\mathbf{n}}$ exactly. $\checkmark$

Validity (non-overlap of adjacent arcs). The two set-backs that share edge $k-1$ are ℓ_{k-1} (from vertex $k-1$) and ℓ_k (from vertex k); the flat part survives iff $\ell_{k-1} + \ell_k < L_{k-1}$. With equilateral edges $L_k \equiv L$ a sufficient condition for all corners is

$$2\delta \tan \frac{\phi_k}{2} < L \quad \forall k \implies \delta < \frac{L}{2 \tan(\pi/N)} \quad \text{for a regular } N\text{-gon } (\phi_k = 2\pi/N). \quad (3.7)$$

The companion estimate $\tan(\pi/N) \approx \pi/N$ holds only as $N \rightarrow \infty$ and is not used as an equality; the exact inequality is checked.

3.3 Arc span

The arc runs from $\mathbf{a}^-$ to $\mathbf{a}^+$ about $\mathbf{z}$. Its starting angle (of $\mathbf{a}^- - \mathbf{z}$ from the $+x$ axis) is

$$\varphi_0 = \text{atan2}(a_y^- - z_y, a_x^- - z_x). \quad (3.8)$$

Theorem 3 (The arc subtends the turning angle). The arc sweep equals the exterior angle, $\Delta\phi = \phi$.

Proof. The radius to $\mathbf{a}^-$ is $\mathbf{z} - \mathbf{a}^- = \delta \hat{\mathbf{n}}$, i.e. along the inward normal of the incoming edge; the radius to $\mathbf{a}^+$ is, by the symmetric tangency argument, $\mathbf{z} - \mathbf{a}^+ = \delta \hat{\mathbf{v}}^\perp$ (the inward normal of the outgoing edge). The angle between two unit vectors equals the angle between their 90° rotations, so the angle between the radii equals the angle between $\hat{\mathbf{u}}$ and $\hat{\mathbf{v}}$, which is the turning angle ϕ . The arc therefore subtends exactly $\Delta\phi = \phi$. $\square$

3.4 Straight section

The flat part of edge k joins $\mathbf{a}_k^+$ to $\mathbf{a}_{k+1}^-$; it is a straight segment of length

$$L_k - \ell_k - \ell_{k+1}. \quad (3.9)$$

3.5 Area of the rounded polygon

The rounded area is the backbone area minus the material sliced from each corner,

$$A_{\bar{P}} = A_{\text{bb}} - \sum_{k=0}^{N-1} \Delta A_k, \quad (3.10)$$

where ΔA_k is bounded by the two tangent segments $\mathbf{a}^- \rightarrow \mathbf{a} \rightarrow \mathbf{a}^+$ and the arc $\mathbf{a}^+ \rightarrow \mathbf{a}^-$. We compute it two independent ways and check they agree.

Method 1: triangle minus circular segment. Triangle $\mathbf{a}^- \mathbf{a} \mathbf{a}^+$ has two sides of length ℓ meeting at the *interior* angle of the corner, which is $\pi - \phi$ (the interior angle is the supplement of the exterior turning angle). Its area is

$$T = \frac{1}{2} \ell^2 \sin(\pi - \phi) = \frac{1}{2} \ell^2 \sin \phi. \quad (3.11)$$

The rounded body keeps the part of that triangle on the far side of the arc; the piece removed beyond the chord $\mathbf{a}^- \mathbf{a}^+$ is the circular segment between the chord and the arc, of area (sector minus the isocles triangle $\mathbf{z} \mathbf{a}^- \mathbf{a}^+$ of apex angle $\Delta\phi = \phi$)

$$S = \frac{1}{2} \delta^2 \phi - \frac{1}{2} \delta^2 \sin \phi = \frac{1}{2} \delta^2 (\phi - \sin \phi). \quad (3.12)$$

The sliced corner is the triangle with the segment *returned* to the body: $\Delta A = T - S$.

Method 2: kite minus sector. The quadrilateral $\mathbf{a}^- \mathbf{a} \mathbf{a}^+ \mathbf{z}$ is a kite: two right triangles $\mathbf{a} \mathbf{a}^- \mathbf{z}$ and $\mathbf{a} \mathbf{a}^+ \mathbf{z}$, each right-angled at the tangent point (radius $\perp$ tangent), with legs ℓ and δ . Its area is $2 \cdot \frac{1}{2} \ell \delta = \ell \delta$. Subtracting the circular sector $\mathbf{z} \mathbf{a}^- \mathbf{a}^+$ of area $\frac{1}{2} \delta^2 \phi$ gives $\Delta A = \ell \delta - \frac{1}{2} \delta^2 \phi$.

Agreement and closed form. With $\ell = \delta \tan(\phi/2)$, write everything in δ, ϕ . Method 2 gives directly

$$\Delta A = \ell \delta - \frac{1}{2} \delta^2 \phi = \delta^2 \tan \frac{\phi}{2} - \frac{1}{2} \delta^2 \phi = \delta^2 \left[\tan \frac{\phi}{2} - \frac{\phi}{2} \right]. \quad (3.13)$$

For Method 1, substitute $\ell^2 = \delta^2 \tan^2(\phi/2)$ and use $\tan^2(\phi/2) \sin \phi = \frac{\sin^2(\phi/2)}{\cos^2(\phi/2)} \cdot 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} = 2 \frac{\sin^3(\phi/2)}{\cos(\phi/2)}$ and $\sin \phi = 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2}$:

$$T - S = \frac{1}{2} \delta^2 \left[2 \frac{\sin^3(\phi/2)}{\cos(\phi/2)} + 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2} \right] - \frac{1}{2} \delta^2 \phi = \delta^2 \frac{\sin \frac{\phi}{2}}{\cos \frac{\phi}{2}} \left(\sin^2 \frac{\phi}{2} + \cos^2 \frac{\phi}{2} \right) - \frac{1}{2} \delta^2 \phi = \delta^2 \tan \frac{\phi}{2} - \frac{1}{2} \delta^2 \phi,$$

identical to Method 2.

$$\Delta A_k = \frac{1}{2} \ell_k^2 \sin \phi_k - \frac{1}{2} \delta^2 (\phi_k - \sin \phi_k) = \ell_k \delta - \frac{1}{2} \delta^2 \phi_k = \delta^2 \left[\tan \frac{\phi_k}{2} - \frac{\phi_k}{2} \right] \geq 0, \quad (3.14)$$

which vanishes as $\delta^2 \phi_k^3/24$ for small turns (Taylor: $\tan x - x = x^3/3 + \dots$ at $x = \phi_k/2$, so $\delta^2 (\phi_k/2)^3/3 = \delta^2 \phi_k^3/24$).

4 Overlap area as a soft potential

For two rounded polygons $\tilde{A}, \tilde{B}$ define the pair energy as their overlap area,

$$E_{AB} = \text{Area}(\tilde{A} \cap \tilde{B}), \quad E = \sum_{A < B} E_{AB}. \quad (4.1)$$

$E = 0$ for disjoint bodies and grows with interpenetration, so $-\nabla E$ is a repulsion that vanishes at contact: the soft-disk potential, generalised to rounded polygons.

4.1 Per-feature locality

Theorem 4 (Boundary localisation). $\text{Area}(\tilde{A} \cap \tilde{B})$ equals the sum, over all boundary features of $\tilde{A}$, of the Green integral (1) taken only over the sub-arcs of each feature that lie inside $\tilde{B}$, plus the symmetric sum with A, B exchanged. Each family is traversed in its native CCW orientation; no extra factor of $\frac{1}{2}$ or sign flip arises, because the $\frac{1}{2}$ already sits inside the Green integrand.

Proof idea. $\partial(\tilde{A} \cap \tilde{B})$ consists of arcs of $\partial\tilde{A}$ that lie inside $\tilde{B}$ and arcs of $\partial\tilde{B}$ that lie inside $\tilde{A}$ (transversal intersection; tangencies are measure zero). Apply Theorem 1 to $\tilde{A} \cap \tilde{B}$: the contour integral decomposes into these two families. The pieces of $\partial\tilde{A}$ outside $\tilde{B}$ are not on $\partial(\tilde{A} \cap \tilde{B})$ and do not contribute. Hence one may walk each feature of $\partial\tilde{A}$, clip it against $\tilde{B}$, and integrate only the inside sub-intervals; the $\partial\tilde{B}$ -inside- $\tilde{A}$ family is recovered when the roles of A and B are swapped. The interior of $\tilde{A} \cap \tilde{B}$ lies on the same side of each bounding arc as the interior of its parent body, so every arc keeps its native CCW sense and the two families simply add (the kernel accumulates both with `atomicAdd`), giving the full area with no extra factor. $\square$

The practical content: never construct the clipped polygon. For each feature (edge E_i or arc K_i) of $\tilde{A}$, find the parameter sub-intervals inside $\tilde{B}$ and add their signed Green contribution. We now derive those two contributions in closed form.

4.2 Contribution of a straight feature

Let an inside sub-interval of a straight feature have endpoints $\mathbf{X}_0, \mathbf{X}_1$, and parametrise the segment between them by $\mathbf{X}(s) = (1-s)\mathbf{X}_0 + s\mathbf{X}_1$, $s \in [0, 1]$. By exactly the calculation (2.3) (with $\mathbf{a}_k \rightarrow \mathbf{X}_0$, $\mathbf{a}_{k+1} \rightarrow \mathbf{X}_1$),

$$\int_0^1 \epsilon_{ij} X^i dX^j = \epsilon_{ij} X_0^i X_1^j, \quad (4.2)$$

and the Green area contribution is half of this:

$$\delta A_{\text{edge}}[\mathbf{X}_0, \mathbf{X}_1] = \frac{1}{2} \epsilon_{ij} X_0^i X_1^j = \frac{1}{2} (X_0^x X_1^y - X_0^y X_1^x). \quad (4.3)$$

This matches the kernel line `localEnergy += 0.5*(X0x*X1y - X0y*X1x)`.

4.3 Contribution of an arc feature

An arc of radius δ about centre $\mathbf{z}$ is $\mathbf{X}(\varphi) = \mathbf{z} + \delta(\cos \varphi, \sin \varphi)$, so

$$dx = -\delta \sin \varphi d\varphi, \quad dy = +\delta \cos \varphi d\varphi. \quad (4.4)$$

The Green integrand is

$$\begin{aligned} \epsilon_{ij} X^i dX^j &= x dy - y dx = (z_x + \delta \cos \varphi)(\delta \cos \varphi) d\varphi - (z_y + \delta \sin \varphi)(-\delta \sin \varphi) d\varphi \\ &= \delta(z_x \cos \varphi + z_y \sin \varphi) d\varphi + \delta^2(\cos^2 \varphi + \sin^2 \varphi) d\varphi \\ &= \delta(z_x \cos \varphi + z_y \sin \varphi) d\varphi + \delta^2 d\varphi. \end{aligned} \quad (4.5)$$

Integrate from φ_0 to φ_1 using $\int \cos = \sin$, $\int \sin = -\cos$:

$$\int_{\varphi_0}^{\varphi_1} (x dy - y dx) = \delta z_x (\sin \varphi_1 - \sin \varphi_0) - \delta z_y (\cos \varphi_1 - \cos \varphi_0) + \delta^2 \Delta\psi, \quad \Delta\psi := \varphi_1 - \varphi_0. \quad (4.6)$$

Now use the endpoint identities $\delta \cos \varphi_{0/1} = X_{0/1}^x - z_x$ and $\delta \sin \varphi_{0/1} = X_{0/1}^y - z_y$:

$$\begin{aligned} \delta z_x (\sin \varphi_1 - \sin \varphi_0) &= z_x [(X_1^y - z_y) - (X_0^y - z_y)] = z_x (X_1^y - X_0^y), \\ -\delta z_y (\cos \varphi_1 - \cos \varphi_0) &= -z_y [(X_1^x - z_x) - (X_0^x - z_x)] = -z_y (X_1^x - X_0^x). \end{aligned} \quad (4.7)$$

Therefore

$$\int_{\varphi_0}^{\varphi_1} (x dy - y dx) = z_x (X_1^y - X_0^y) - z_y (X_1^x - X_0^x) + \delta^2 \Delta\psi. \quad (4.8)$$

Reduction to the kernel form. We now show (4.8) is precisely a chord (origin-triangle) term plus a circular-segment term. Expand the cross product of the endpoints directly:

$$\begin{aligned} \epsilon_{ij} X_0^i X_1^j &= (z_x + \delta \cos \varphi_0)(z_y + \delta \sin \varphi_1) - (z_y + \delta \sin \varphi_0)(z_x + \delta \cos \varphi_1) \\ &= \delta z_x (\sin \varphi_1 - \sin \varphi_0) - \delta z_y (\cos \varphi_1 - \cos \varphi_0) + \delta^2 (\cos \varphi_0 \sin \varphi_1 - \sin \varphi_0 \cos \varphi_1) \\ &= [z_x (X_1^y - X_0^y) - z_y (X_1^x - X_0^x)] + \delta^2 \sin \Delta\psi, \end{aligned} \quad (4.9)$$

where the last line used the two endpoint identities again and $\cos \varphi_0 \sin \varphi_1 - \sin \varphi_0 \cos \varphi_1 = \sin(\varphi_1 - \varphi_0) = \sin \Delta\psi$. Solving (4.9) for the bracket and substituting into (4.8),

$$\int_{\varphi_0}^{\varphi_1} (x dy - y dx) = \epsilon_{ij} X_0^i X_1^j - \delta^2 \sin \Delta\psi + \delta^2 \Delta\psi. \quad (4.10)$$

Halving gives the area contribution.

$$\delta A_{\text{arc}}[\mathbf{X}_0, \mathbf{X}_1; \Delta\psi] = \frac{1}{2} \epsilon_{ij} X_0^i X_1^j + \frac{1}{2} \delta^2 (\Delta\psi - \sin \Delta\psi). \quad (4.11)$$

The first term is the signed area of the origin triangle $\mathbf{0} \mathbf{X}_0 \mathbf{X}_1$ (the chord), the second is the circular segment between chord and arc. This matches `localEnergy += 0.5*(X0x*X1y - X0y*X1x) + 0.5*delta*`

Sanity check

Take a quarter circle of unit radius centred at the origin: $\mathbf{X}_0 = (1, 0)$, $\mathbf{X}_1 = (0, 1)$, $\Delta\psi = \pi/2$, $\delta = 1$. (4.11) gives $\frac{1}{2}(1 \cdot 1 - 0) + \frac{1}{2}(\pi/2 - 1) = \frac{1}{2} + \pi/4 - \frac{1}{2} = \pi/4$, the exact quarter-disk area. The general formula (4.8) gives the same value, since with $\mathbf{z} = \mathbf{0}$ it reduces to $\frac{1}{2}\delta^2\Delta\psi = \pi/4$.

4.4 Point in the smoothed polygon

Membership in $\tilde{B}$ is decided by a horizontal ray cast from a query point $\mathbf{p}$ toward $+x$: count transversal crossings of $\partial\tilde{B}$; an odd count means $\mathbf{p} \in \tilde{B}$ (Jordan curve parity). The midpoint of each candidate feature sub-interval is tested.

Straight feature $\mathbf{b}_j^+ \rightarrow \mathbf{b}_{j+1}^-$: the standard ray-segment test — the segment is crossed iff its endpoints straddle $y = p_y$ and the x of the crossing exceeds p_x .

Arc feature centre (z_x, z_y) , radius δ : the ray $y = p_y$ meets the circle where $(x - z_x)^2 + (p_y - z_y)^2 = \delta^2$, i.e.

$$x = z_x \pm \sqrt{\delta^2 - (p_y - z_y)^2} \quad (\text{real iff } |p_y - z_y| \leq \delta). \quad (4.12)$$

Each real root is kept iff it lies on the arc's angular range, $\text{atan2}(p_y - z_y, x - z_x) \in [\varphi_0, \varphi_0 + \Delta\phi]$, and satisfies $x > p_x$.

4.5 Feature-feature intersections

Clipping a feature against $\tilde{B}$ requires the parameters at which it crosses each feature of $\partial\tilde{B}$.

Arc-edge. Edge $\mathbf{r}(t) = \mathbf{p} + t\mathbf{e}$, $t \in (0, 1)$; circle $|\mathbf{r} - \mathbf{z}|^2 = \delta^2$. Put $\mathbf{f} = \mathbf{p} - \mathbf{z}$. Then $|\mathbf{f} + t\mathbf{e}|^2 = \delta^2$ expands to the quadratic

$$\underbrace{|\mathbf{e}|^2}_a t^2 + \underbrace{2(\mathbf{f} \cdot \mathbf{e})}_b t + \underbrace{|\mathbf{f}|^2 - \delta^2}_c = 0, \quad t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (b^2 \geq 4ac), \quad (4.13)$$

keeping roots with $t \in (0, 1)$ whose point lies on the arc's angular range.

Arc-arc. Circles $(\mathbf{z}_A, \delta_A)$ and $(\mathbf{z}_B, \delta_B)$; set $d = |\mathbf{z}_B - \mathbf{z}_A|$ and $\hat{\mathbf{s}} = (\mathbf{z}_B - \mathbf{z}_A)/d$. Write a common point as $\mathbf{X} = \mathbf{z}_A + a_2 \hat{\mathbf{s}} + h \hat{\mathbf{s}}^\perp$. Membership of both circles requires

$$\delta_A^2 = a_2^2 + h^2, \quad \delta_B^2 = (d - a_2)^2 + h^2. \quad (4.14)$$

Subtract to eliminate h : $\delta_A^2 - \delta_B^2 = a_2^2 - (d - a_2)^2 = 2da_2 - d^2$, hence

$$a_2 = \frac{d^2 + \delta_A^2 - \delta_B^2}{2d}, \quad h = \sqrt{\max(0, \delta_A^2 - a_2^2)}, \quad \mathbf{X} = \mathbf{z}_A + a_2 \hat{\mathbf{s}} \pm h \hat{\mathbf{s}}^\perp. \quad (4.15)$$

Each candidate is checked against both arcs' angular ranges.

Edge-edge. Lines $\mathbf{p} + t\mathbf{d}$ and $\mathbf{r} + s\mathbf{e}$. Equate and rearrange: $t\mathbf{d} - s\mathbf{e} = \mathbf{r} - \mathbf{p}$. Cross (the scalar ϵ -product) the equation with $\mathbf{e}$, then with $\mathbf{d}$, using $\mathbf{w} \times \mathbf{w} = 0$:

$$t(\mathbf{d} \times \mathbf{e}) = (\mathbf{r} - \mathbf{p}) \times \mathbf{e}, \quad -s(\mathbf{e} \times \mathbf{d}) = (\mathbf{r} - \mathbf{p}) \times \mathbf{d}, \quad (4.16)$$

and with $\mathbf{e} \times \mathbf{d} = -(\mathbf{d} \times \mathbf{e})$,

$$t = \frac{(\mathbf{r} - \mathbf{p}) \times \mathbf{e}}{\mathbf{d} \times \mathbf{e}}, \quad s = \frac{(\mathbf{r} - \mathbf{p}) \times \mathbf{d}}{\mathbf{d} \times \mathbf{e}}. \quad (4.17)$$

A proper crossing requires $t, s \in (0, 1)$ and $\mathbf{d} \times \mathbf{e} \neq 0$.

5 The energy in full

Everything derived so far now collapses into a single closed expression for the total energy E as a function of nothing but the raw backbone vertex coordinates $\{\mathbf{a}_k^P\}$ and the per-polygon rounding radii $\{\delta_P\}$. We state the master equation first, then define, in cascading layers, every subfunction that appears in it — right down to the vertices.

5.1 Layer 0: per-corner geometry operator

For a polygon P define, at each vertex k , the geometry operator $\mathcal{G}_P(k)$ that returns the sextuple $(\ell_k^P, \mathbf{a}_k^{P,-}, \mathbf{a}_k^{P,+}, \mathbf{z}_k^P, \varphi_{0,k}^P, \Delta\phi_k^P)$ via

$$\boxed{\begin{aligned} \mathbf{u} &= \text{mic}(\mathbf{a}_k^P - \mathbf{a}_{k-1}^P), & \mathbf{v} &= \text{mic}(\mathbf{a}_{k+1}^P - \mathbf{a}_k^P), & \hat{\mathbf{u}} &= \frac{\mathbf{u}}{|\mathbf{u}|}, & \hat{\mathbf{v}} &= \frac{\mathbf{v}}{|\mathbf{v}|}, \\ c &= \epsilon_{ij} \hat{u}^i \hat{v}^j, & d &= \delta_{ij} \hat{u}^i \hat{v}^j, & \Delta\phi_k^P &= \phi = \text{atan2}(c, d), \\ \ell_k^P &= \frac{\delta_P c}{1+d} = \delta_P \tan \frac{\phi}{2}, & \hat{\mathbf{n}} &= \hat{\mathbf{u}}^\perp = (-\hat{u}^2, \hat{u}^1), \\ \mathbf{a}_k^{P,-} &= \mathbf{a}_k^P - \ell_k^P \hat{\mathbf{u}}, & \mathbf{a}_k^{P,+} &= \mathbf{a}_k^P + \ell_k^P \hat{\mathbf{v}}, & \mathbf{z}_k^P &= \mathbf{a}_k^P - \ell_k^P \hat{\mathbf{u}} + \delta_P \hat{\mathbf{n}}, \\ \varphi_{0,k}^P &= \text{atan2}(\mathbf{a}_{k,y}^{P,-} - \mathbf{z}_{k,y}^P, \mathbf{a}_{k,x}^{P,-} - \mathbf{z}_{k,x}^P). \end{aligned}} \quad (5.1)$$

5.2 Layer 1: feature parametrisations

The boundary $\partial\tilde{P}$ is the cyclic alternation $K_0^P, E_0^P, \dots, K_{N_P-1}^P, E_{N_P-1}^P$, with the arc at vertex i and the flat section of edge i given by

$$K_i^P(\tau) = \mathbf{z}_i^P + \delta_P \left(\cos(\varphi_{0,i}^P + \tau \Delta\phi_i^P), \sin(\varphi_{0,i}^P + \tau \Delta\phi_i^P) \right), \quad \tau \in [0, 1], \quad (5.2)$$

$$E_i^P(s) = (1-s) \mathbf{a}_i^{P,+} + s \mathbf{a}_{i+1}^{P,-}, \quad s \in [0, 1]. \quad (5.3)$$

5.3 Layer 2: the two Green building blocks

For any ordered endpoint pair $\mathbf{X}, \mathbf{Y}$ and any arc sweep $\Delta\psi$, define the chord (origin-triangle) functional and the circular-cap functional,

$$\boxed{\mathcal{S}(\mathbf{X}, \mathbf{Y}) := \frac{1}{2} \epsilon_{ij} X^i Y^j = \frac{1}{2} (X^x Y^y - X^y Y^x), \quad \mathcal{Q}_\delta(\Delta\psi) := \frac{1}{2} \delta^2 (\Delta\psi - \sin \Delta\psi).} \quad (5.4)$$

By (4.3) and (4.11), the Green contribution of an inside sub-interval is $\mathcal{S}$ alone for a straight feature, and $\mathcal{S} + \mathcal{Q}_{\delta_P}$ for an arc feature.

5.4 Layer 3: clipping — the inside-interval set

The ray-cast parity test for a point $\mathbf{p}$ against the rounded body $\tilde{Q}$ sums one indicator per feature of $\partial\tilde{Q}$,

$$\Pi_{\tilde{Q}}(\mathbf{p}) = \left[\sum_j \chi_j^E(\mathbf{p}) + \sum_j \chi_j^K(\mathbf{p}) \right] \bmod 2 \in \{0, 1\}, \quad (5.5)$$

where, for the rightward ray $\{(x, p_y) : x > p_x\}$,

$$\chi_j^E(\mathbf{p}) = \mathbb{K} \left[(b_{j,y}^{Q,+} - p_y)(b_{j+1,y}^{Q,-} - p_y) < 0 \wedge x_{\times} > p_x \right], \quad x_{\times} = \text{ray} \cap \text{segment } x, \quad (5.6)$$

$$\chi_j^K(\mathbf{p}) = \# \left\{ x = z_{j,x}^Q \pm \sqrt{\delta_Q^2 - (p_y - z_{j,y}^Q)^2} : x > p_x, \right. \\ \left. \text{atan2}(p_y - z_{j,y}^Q, x - z_{j,x}^Q) \in [\varphi_{0,j}^Q, \varphi_{0,j}^Q + \Delta\phi_j^Q] \right\}. \quad (5.7)$$

The crossing-parameter set of a feature F of $\partial\tilde{P}$ against $\tilde{Q}$ is the union of the feature–feature intersection parameters (arc–edge quadratic, arc–arc radical axis, edge–edge determinant; §5.7),

$$\mathcal{T}(F; \tilde{Q}) = \bigcup_{G \in \partial\tilde{Q}} \{t \in (0, 1) : F(t) = G(t') \text{ for some admissible } t'\}. \quad (5.8)$$

Sorting $\{0\} \cup \mathcal{T}(F; \tilde{Q}) \cup \{1\}$ into $0 = \xi_0 < \xi_1 < \dots < \xi_{m+1} = 1$, the inside-interval set keeps exactly the bands whose midpoint is interior:

$$\mathcal{I}(F; \tilde{Q}) = \left\{ [\xi_r, \xi_{r+1}] : \Pi_{\tilde{Q}}(F(\frac{\xi_r + \xi_{r+1}}{2})) = 1, \quad 0 \leq r \leq m \right\}. \quad (5.9)$$

5.5 Layer 4: periodic placement

Within the pair (A, B) , body B is rigidly placed in the single periodic image nearest A : anchoring on the reference vertex $\mathbf{a}_0^A$,

$$\tilde{\mathbf{b}}_0 = \mathbf{a}_0^A + \text{mic}(\mathbf{b}_0^B - \mathbf{a}_0^A), \quad \tilde{\mathbf{b}}_j = \tilde{\mathbf{b}}_0 + (\mathbf{b}_j^B - \mathbf{b}_0^B), \quad (5.10)$$

and $\tilde{B}$ in everything below is built from $\{\tilde{\mathbf{b}}_j\}$ via (5.1)–(5.3). (The kernel re-anchors per interacting vertex; since a polygon is small against the box, the placed copy is the same.)

5.6 The master equation

Writing (P, Q) for the two directed roles of an interacting pair, the total energy is the pair sum of, for each polygon's every feature, the Green building blocks evaluated on the endpoints of each inside sub-interval:

$$E = \sum_{A < B} \sum_{(P,Q) \in \{(A,B), (B,A)\}} \sum_{i=0}^{N_P-1} \left[\begin{aligned} & \sum_{[s_0, s_1] \in \mathcal{I}(E_i^P; \tilde{Q})} \mathcal{S}(E_i^P(s_0), E_i^P(s_1)) \\ & + \sum_{[\tau_0, \tau_1] \in \mathcal{I}(K_i^P; \tilde{Q})} \left(\mathcal{S}(K_i^P(\tau_0), K_i^P(\tau_1)) \right. \\ & \quad \left. + \mathcal{Q}_{\delta_P}((\tau_1 - \tau_0) \Delta\phi_i^P) \right) \end{aligned} \right] \quad (5.11)$$

with the dependency chain, read inward:

$$E \leftarrow \mathcal{S}, \mathcal{Q} \leftarrow E_i^P, K_i^P \leftarrow \mathcal{I}(\leftarrow \Pi, \mathcal{T}) \leftarrow \mathcal{G}_P \leftarrow \{\mathbf{a}_k^P\}, \delta_P, \text{ and the placement (5.10).}$$

How to read (5.11)

The three outer sums are: over interacting pairs; over the two directions (features of $\tilde{A}$ clipped to $\tilde{B}$, then $\tilde{B}$ to $\tilde{A}$ — Theorem 4); and over the N_P vertices of the current polygon. For each vertex two features are emitted: the arc K_i^P and the following flat section E_i^P . Each is clipped to its inside-interval set $\mathcal{I}$ (5.9); every surviving band contributes its chord term $\mathcal{S}$, and arcs add the cap $\mathcal{Q}$. Unrolling $\mathcal{S}, \mathcal{Q}$ via (5.4), the feature parametrisations via (5.2)–(5.3), and the arc data via the geometry operator (5.1) expresses E purely in the vertex coordinates — the “gory” form. Nothing else enters: no clipped polygon is ever constructed.

5.7 Fully unrolled arc term

To exhibit the depth concretely, take a single inside band $[\tau_0, \tau_1]$ of an arc K_i^P with the layer-0 abbreviations $\varphi_r = \varphi_{0,i}^P + \tau_r \Delta \phi_i^P$ ($r = 0, 1$) and $\Delta \psi = (\tau_1 - \tau_0) \Delta \phi_i^P$. Its contribution, with *everything* substituted, is

$$\frac{1}{2} \epsilon_{mn} \left[z_i^P + \delta_P (\cos \varphi_0, \sin \varphi_0) \right]^m \left[z_i^P + \delta_P (\cos \varphi_1, \sin \varphi_1) \right]^n + \frac{1}{2} \delta_P^2 (\Delta \psi - \sin \Delta \psi), \quad (5.12)$$

in which $z_i^P = \mathbf{a}_i^P - \ell_i^P \hat{\mathbf{u}} + \delta_P \hat{\mathbf{u}}^\perp$, $\ell_i^P = \delta_P \tan(\phi_i^P/2)$, $\phi_i^P = \text{atan2}(\epsilon_{ab} \hat{u}^a \hat{v}^b, \hat{u}^a \hat{v}^a)$, and $\hat{\mathbf{u}}, \hat{\mathbf{v}}$ are the normalised minimum-image differences of $\mathbf{a}_{i-1}^P, \mathbf{a}_i^P, \mathbf{a}_{i+1}^P$ from (5.1). The straight-feature band is the same with the cap dropped and the endpoints taken from (5.3).

6 The analytic force

The force on vertex $\mathbf{a}_\alpha$ is $\mathbf{F}_\alpha = -\partial E / \partial \mathbf{a}_\alpha$. Because the inside sub-intervals of every feature move as the vertices move (their endpoints are crossing points that depend on *both* polygons), the energy is an integral with parameter-dependent integrand *and* parameter-dependent limits. Its derivative needs the Leibniz rule.

6.1 Leibniz rule for a signed-area contribution

Lemma 5 (Differentiating a moving signed-area integral). Let a feature be $\gamma(t; \theta)$ with θ a vertex coordinate, and let the inside sub-interval contribute $I(\theta) = \frac{1}{2} \int_{t_0(\theta)}^{t_1(\theta)} \epsilon_{ij} \gamma^i \dot{\gamma}^j dt$ with $\dot{\gamma} = \partial_t \gamma$. Then

$$\frac{\partial I}{\partial \theta} = \underbrace{\frac{1}{2} \int_{t_0}^{t_1} \epsilon_{ij} \frac{\partial}{\partial \theta} (\gamma^i \dot{\gamma}^j) dt}_{\text{direct term}} + \underbrace{\left[\frac{1}{2} \epsilon_{ij} \gamma^i \dot{\gamma}^j \right]_{t_1} \frac{\partial t_1}{\partial \theta} - \left[\frac{1}{2} \epsilon_{ij} \gamma^i \dot{\gamma}^j \right]_{t_0} \frac{\partial t_0}{\partial \theta}}_{\text{boundary (limit) term}}. \quad (6.1)$$

Proof. This is the Leibniz integral rule $\frac{d}{d\theta} \int_{t_0(\theta)}^{t_1(\theta)} F(t, \theta) dt = \int_{t_0}^{t_1} \partial_\theta F dt + F(t_1, \theta) t_1' - F(t_0, \theta) t_0'$ applied to $F = \frac{1}{2} \epsilon_{ij} \gamma^i \dot{\gamma}^j$. $\square$

The direct term is the curve deforming with limits held fixed; the boundary term is the in/out crossing points sliding along a fixed curve.

6.2 The implicit-function rule for crossing parameters

Every crossing parameter t_* is defined by a scalar equation $\Phi(t_*; \text{vertices}) = 0$ (e.g. “the point lies on the other feature”). By the implicit function theorem,

$$\boxed{\frac{\partial t_*}{\partial \theta} = - \frac{\partial \Phi / \partial \theta}{\partial \Phi / \partial t_*} \Big|_{t_*}}. \quad (6.2)$$

This single formula generates every boundary term below.

6.3 Force from a straight feature

For the flat section E_i from $\mathbf{P} = \mathbf{a}_i^+$ to $\mathbf{Q} = \mathbf{a}_{i+1}^-$, parametrise $\gamma(t) = (1-t)\mathbf{P} + t\mathbf{Q}$, so $\dot{\gamma} = \mathbf{Q} - \mathbf{P} := \mathbf{e}_A$ and the inside contribution is, by (4.3), $I = \frac{1}{2} \epsilon_{ij} X_0^i X_1^j$ with $\mathbf{X}_r = \gamma(t_r)$, $r = 0, 1$.

Direct term. Holding t_0, t_1 fixed, $\partial \mathbf{X}_r / \partial \mathbf{P} = (1-t_r) \mathbb{K}$ and $\partial \mathbf{X}_r / \partial \mathbf{Q} = t_r \mathbb{K}$ (the barycentric weights). Hence

$$\begin{aligned} \frac{\partial I}{\partial P^m} \Big|_{\text{dir}} &= \frac{1}{2} \epsilon_{ij} \left[\frac{\partial X_0^i}{\partial P^m} X_1^j + X_0^i \frac{\partial X_1^j}{\partial P^m} \right] = \frac{1}{2} \epsilon_{ij} \left[(1-t_0) \delta_m^i X_1^j + X_0^i (1-t_1) \delta_m^j \right] \\ &= (1-t_0) \underbrace{\frac{1}{2} \epsilon_{mj} X_1^j}_{:= f_0^m} + (1-t_1) \underbrace{\frac{1}{2} \epsilon_{im} X_0^i}_{:= f_1^m}, \end{aligned} \quad (6.3)$$

with $\mathbf{f}_0 = \frac{1}{2} (X_1^y, -X_1^x)$ and $\mathbf{f}_1 = \frac{1}{2} (-X_0^y, X_0^x)$. By the same computation with the $\mathbf{Q}$ -weights,

$$\frac{\partial I}{\partial \mathbf{P}} \Big|_{\text{dir}} = (1-t_0) \mathbf{f}_0 + (1-t_1) \mathbf{f}_1, \quad \frac{\partial I}{\partial \mathbf{Q}} \Big|_{\text{dir}} = t_0 \mathbf{f}_0 + t_1 \mathbf{f}_1. \quad (6.4)$$

Boundary term. Suppose the inside interval begins where E_i crosses a straight feature of $\tilde{B}$, the line through $\mathbf{R}$ with direction $\mathbf{e}_B$. The crossing parameter solves $\Phi(t) = (\gamma(t) - \mathbf{R}) \times \mathbf{e}_B = 0$, i.e. $\Phi(t) = \epsilon_{ij} ((1-t)\mathbf{P} + t\mathbf{Q} - \mathbf{R})^i e_B^j$. Then

$$\begin{aligned} \frac{\partial \Phi}{\partial t} &= \epsilon_{ij} e_A^i e_B^j = \mathbf{e}_A \times \mathbf{e}_B, \\ \frac{\partial \Phi}{\partial P^m} &= \epsilon_{ij} (1-t) \delta_m^i e_B^j = (1-t) (e_B^\perp)_m, \quad \frac{\partial \Phi}{\partial Q^m} = \epsilon_{ij} t \delta_m^i e_B^j = t (e_B^\perp)_m, \end{aligned} \quad (6.5)$$

where $(e_B^\perp)_m = \epsilon_{mj} e_B^j$. The implicit rule (6.2) gives

$$\frac{\partial t_0}{\partial \mathbf{P}} = \frac{(t_0 - 1) \mathbf{e}_B^\perp}{\mathbf{e}_A \times \mathbf{e}_B}, \quad \frac{\partial t_0}{\partial \mathbf{Q}} = \frac{-t_0 \mathbf{e}_B^\perp}{\mathbf{e}_A \times \mathbf{e}_B}, \quad (6.6)$$

and similarly for t_1 against whichever feature bounds the other end. The matching $\partial I / \partial t_0$ is read from (4.3): since $\partial \mathbf{X}_0 / \partial t_0 = \dot{\gamma} = \mathbf{e}_A$,

$$\frac{\partial I}{\partial t_0} = \frac{1}{2} \epsilon_{ij} \frac{\partial X_0^i}{\partial t_0} X_1^j = \frac{1}{2} \epsilon_{ij} e_A^i X_1^j = \frac{1}{2} (\mathbf{e}_A \times \mathbf{X}_1), \quad \frac{\partial I}{\partial t_1} = -\frac{1}{2} (\mathbf{e}_A \times \mathbf{X}_0). \quad (6.7)$$

The boundary contribution to the force is $\frac{\partial I}{\partial t_0} \frac{\partial t_0}{\partial \mathbf{P}} + \frac{\partial I}{\partial t_1} \frac{\partial t_1}{\partial \mathbf{P}}$ (the “ β -correction” in the kernel), added to the direct term. The total $\partial I / \partial \mathbf{P}$ is then propagated to backbone vertices through the arc-geometry chain rule, because $\mathbf{P} = \mathbf{a}_i^+$ and $\mathbf{Q} = \mathbf{a}_{i+1}^-$ are tangent points, not backbone vertices.

6.4 Force from an arc feature

For arc K_i the inside contribution is, from (4.11),

$$I = \frac{1}{2} \epsilon_{ij} X_0^i X_1^j + \frac{1}{2} \delta^2 (\Delta\psi - \sin \Delta\psi), \quad \Delta\psi = (\tau_1 - \tau_0) \Delta\phi_i, \quad (6.8)$$

with $\mathbf{X}_r = \mathbf{z}_i + \delta(\cos \varphi_r, \sin \varphi_r)$ and $\varphi_r = \varphi_{0,i} + \tau_r \Delta\phi_i$. The arc moves because $\mathbf{z}_i, \varphi_{0,i}, \Delta\phi_i$ each depend on the three backbone vertices $\mathbf{a}_{i-1}, \mathbf{a}_i, \mathbf{a}_{i+1}$. We need the upstream partials of I with respect to the arc's geometric data and then the chain rule back to vertices.

Upstream partials (all-inside case $\tau_0 = 0, \tau_1 = 1$). Then $\mathbf{X}_0 = \mathbf{a}_i^-$, $\mathbf{X}_1 = \mathbf{a}_i^+$ and $\Delta\psi = \Delta\phi_i$. Directly,

$$\begin{aligned} \frac{\partial I}{\partial \mathbf{X}_0} &= \frac{1}{2} \mathbf{X}_1^\perp{}^T \quad \text{i.e.} \quad \frac{\partial I}{\partial X_0^m} = \frac{1}{2} \epsilon_{mj} X_1^j, & \frac{\partial I}{\partial X_1^m} &= -\frac{1}{2} \epsilon_{mj} X_0^j, \\ \frac{\partial I}{\partial (\Delta\psi)} &= \frac{1}{2} \delta^2 (1 - \cos \Delta\psi), & \frac{\partial I}{\partial \delta} &= \delta (\Delta\psi - \sin \Delta\psi). \end{aligned} \quad (6.9)$$

The dependence of $\mathbf{X}_0, \mathbf{X}_1$ on $(\mathbf{z}, \varphi_0, \Delta\phi)$ feeds these into $\partial I / \partial \mathbf{z}$, $\partial I / \partial \varphi_0$, $\partial I / \partial \Delta\phi$ via $\partial \mathbf{X}_r / \partial \mathbf{z} = \mathbb{K}$, $\partial \mathbf{X}_r / \partial \varphi_0 = \delta(-\sin \varphi_r, \cos \varphi_r)$, $\partial \mathbf{X}_r / \partial \Delta\phi = \tau_r \delta(-\sin \varphi_r, \cos \varphi_r)$. For clipped intervals (τ_0, τ_1 themselves crossing parameters) the boundary term of Lemma 5 adds the implicit-rule contribution (6.2), exactly as for edges.

6.5 The arc-geometry chain rule

The routine `arcGeomGrad` receives the upstream sensitivities $\partial E / \partial \mathbf{a}_i^-$, $\partial E / \partial \mathbf{a}_i^+$, $\partial E / \partial \Delta\phi_i$ (and, where the arc term contributes, $\partial E / \partial \mathbf{z}_i$, $\partial E / \partial \varphi_{0,i}$) and returns the forces on $\mathbf{a}_{i-1}, \mathbf{a}_i, \mathbf{a}_{i+1}$. Everything reduces to derivatives of $\ell, \Delta\phi, \mathbf{z}, \mathbf{a}^\pm, \varphi_0$ with respect to $\hat{\mathbf{u}}, \hat{\mathbf{v}}$, and of $\hat{\mathbf{u}}, \hat{\mathbf{v}}$ with respect to the backbone vertices. We derive each block.

Set-back derivatives. With $\ell = \delta c / (1 + d)$, $c = \epsilon_{pq} \hat{u}^p \hat{v}^q$, $d = \hat{u}^p \hat{v}^p$, compute first

$$\frac{\partial c}{\partial \hat{u}^i} = \epsilon_{iq} \hat{v}^q = (\hat{\mathbf{v}}^\perp)_i^-, \quad \frac{\partial c}{\partial \hat{v}^j} = \epsilon_{pj} \hat{u}^p = -(\hat{\mathbf{u}}^\perp)_j^-, \quad \frac{\partial d}{\partial \hat{u}^i} = \hat{v}_i, \quad \frac{\partial d}{\partial \hat{v}^j} = \hat{u}_j, \quad (6.10)$$

where $(\mathbf{w}^\perp)_i^- := \epsilon_{ij} w^j$ (the clockwise rotate; for a convex CCW corner $c > 0$ so no $\text{sgn}(c)$ is needed — in general it carries $\text{sgn}(c)$). The quotient rule then gives

$$\frac{\partial \ell}{\partial \hat{u}^i} = \delta \frac{(\partial_{\hat{u}^i} c)(1 + d) - c(\partial_{\hat{u}^i} d)}{(1 + d)^2} = \frac{1}{1 + d} \left[\delta \epsilon_{iq} \hat{v}^q - \ell \hat{v}_i \right], \quad (6.11)$$

$$\frac{\partial \ell}{\partial \hat{v}^j} = \frac{1}{1 + d} \left[-\delta \epsilon_{jp} \hat{u}^p - \ell \hat{u}_j \right], \quad (6.12)$$

where in (6.11) we used $\ell = \delta c / (1 + d)$ to fold the second term.

Span derivatives. $\Delta\phi = \text{atan2}(c, d)$ with $c^2 + d^2 = 1$, so by Lemma 3 $d\Delta\phi = d\,dc - c\,dd$, hence

$$\frac{\partial \Delta\phi}{\partial \hat{u}^i} = d \epsilon_{iq} \hat{v}^q - c \hat{v}_i, \quad \frac{\partial \Delta\phi}{\partial \hat{v}^j} = -d \epsilon_{jp} \hat{u}^p - c \hat{u}_j. \quad (6.13)$$

Tangent-point and centre derivatives. From $\mathbf{a}^- = \mathbf{a}_i - \ell \hat{\mathbf{u}}$, $\mathbf{a}^+ = \mathbf{a}_i + \ell \hat{\mathbf{v}}$ and $\mathbf{z} = \mathbf{a}_i - \ell \hat{\mathbf{u}} + \delta \hat{\mathbf{n}}$ with $\hat{\mathbf{n}} = \hat{\mathbf{u}}^\perp$,

$$\begin{aligned} \frac{\partial a^{-m}}{\partial \hat{u}^i} &= -\hat{u}^m \frac{\partial \ell}{\partial \hat{u}^i} - \ell \delta_i^m, & \frac{\partial a^{-m}}{\partial \hat{v}^j} &= -\hat{u}^m \frac{\partial \ell}{\partial \hat{v}^j}, \\ \frac{\partial a^{+m}}{\partial \hat{v}^j} &= +\hat{v}^m \frac{\partial \ell}{\partial \hat{v}^j} + \ell \delta_j^m, & \frac{\partial a^{+m}}{\partial \hat{u}^i} &= +\hat{v}^m \frac{\partial \ell}{\partial \hat{u}^i}, \\ \frac{\partial z^m}{\partial \hat{u}^i} &= -\hat{u}^m \frac{\partial \ell}{\partial \hat{u}^i} - \ell \delta_i^m + \delta \frac{\partial (\hat{u}^\perp)^m}{\partial \hat{u}^i}, & \frac{\partial z^m}{\partial \hat{v}^j} &= -\hat{u}^m \frac{\partial \ell}{\partial \hat{v}^j}, \end{aligned} \quad (6.14)$$

with $\partial(\hat{u}^\perp)^m/\partial \hat{u}^i = -\epsilon^m_i$ (the rotation is linear), and $\partial \mathbf{a}^\pm/\partial \mathbf{a}_i = \mathbb{K}$, $\partial \mathbf{z}/\partial \mathbf{a}_i = \mathbb{K}$ for the explicit $\mathbf{a}_i$ dependence.

Unit-vector derivatives. With $\mathbf{u} = \mathbf{a}_i - \mathbf{a}_{i-1}$ and $\mathbf{v} = \mathbf{a}_{i+1} - \mathbf{a}_i$, Lemma 2 gives the projector form

$$\frac{\partial \hat{u}^j}{\partial a_i^k} = \frac{P^{jk}[\hat{\mathbf{u}}]}{|\mathbf{u}|}, \quad \frac{\partial \hat{u}^j}{\partial a_{i-1}^k} = -\frac{P^{jk}[\hat{\mathbf{u}}]}{|\mathbf{u}|}, \quad \frac{\partial \hat{v}^j}{\partial a_{i+1}^k} = \frac{P^{jk}[\hat{\mathbf{v}}]}{|\mathbf{v}|}, \quad \frac{\partial \hat{v}^j}{\partial a_i^k} = -\frac{P^{jk}[\hat{\mathbf{v}}]}{|\mathbf{v}|}, \quad (6.15)$$

with $P^{jk}[\hat{\mathbf{u}}] = \delta^{jk} - \hat{u}^j \hat{u}^k$ and likewise for $\hat{\mathbf{v}}$.

Assembly. The backbone vertex $\mathbf{a}_{i-1}$ enters E only through $\hat{\mathbf{u}}$; $\mathbf{a}_{i+1}$ only through $\hat{\mathbf{v}}$; $\mathbf{a}_i$ enters through both and explicitly through $\mathbf{a}^\pm, \mathbf{z}$. With q ranging over the intermediate quantities $\{\mathbf{a}^-, \mathbf{a}^+, \mathbf{z}, \varphi_0, \Delta\phi\}$, the multivariate chain rule reads

$$\frac{\partial E}{\partial a_{i-1}^k} = \sum_q \frac{\partial E}{\partial q} \frac{\partial q}{\partial \hat{u}^p} \frac{\partial \hat{u}^p}{\partial a_{i-1}^k}, \quad \frac{\partial E}{\partial a_{i+1}^k} = \sum_q \frac{\partial E}{\partial q} \frac{\partial q}{\partial \hat{v}^p} \frac{\partial \hat{v}^p}{\partial a_{i+1}^k}, \quad (6.16)$$

$$\frac{\partial E}{\partial a_i^k} = \left. \frac{\partial E}{\partial q} \right|_{\text{explicit}} + \sum_q \frac{\partial E}{\partial q} \left(\frac{\partial q}{\partial \hat{u}^p} \frac{\partial \hat{u}^p}{\partial a_i^k} + \frac{\partial q}{\partial \hat{v}^p} \frac{\partial \hat{v}^p}{\partial a_i^k} \right). \quad (6.17)$$

Substituting (6.11)–(6.15) and collecting the gradients $\partial E/\partial \mathbf{u} := P[\hat{\mathbf{u}}] \cdot (\partial E/\partial \hat{\mathbf{u}})/|\mathbf{u}|$ (the component perpendicular to $\hat{\mathbf{u}}$, since $|\hat{\mathbf{u}}| = 1$ kills the parallel part), the four nonzero vertex forces take the compact form the kernel uses,

$$\mathbf{f}_{i-1} = -\frac{1}{|\mathbf{u}|} P[\hat{\mathbf{u}}] \cdot \frac{\partial E}{\partial \hat{\mathbf{u}}}, \quad \mathbf{f}_{i+1} = \frac{1}{|\mathbf{v}|} P[\hat{\mathbf{v}}] \cdot \frac{\partial E}{\partial \hat{\mathbf{v}}}, \quad \mathbf{f}_i = \mathbf{f}^{a^-} + \mathbf{f}^{a^+} + \frac{1}{|\mathbf{u}|} P[\hat{\mathbf{u}}] \cdot \frac{\partial E}{\partial \hat{\mathbf{u}}} - \frac{1}{|\mathbf{v}|} P[\hat{\mathbf{v}}] \cdot \frac{\partial E}{\partial \hat{\mathbf{v}}}, \quad (6.18)$$

where $\mathbf{f}^{a^\pm}$ are the explicit- $\mathbf{a}_i$ contributions through the tangent points. The projector P appears automatically: any gradient routed through a unit vector is physically a torque about that edge and cannot have a component that would change the (constrained) edge length — the calculus enforces this through Lemma 2.

The chain rule looks heavy but every block is one of four objects: a quotient-rule derivative of ℓ , an `atan2` derivative of $\Delta\phi$, a linear rotation derivative of $\hat{\mathbf{n}}$, and the projector derivative of a unit vector. Read top-down it is just $E \rightarrow (\mathbf{a}^\pm, \Delta\phi) \rightarrow (\ell, \hat{\mathbf{u}}, \hat{\mathbf{v}}) \rightarrow (\mathbf{u}, \mathbf{v}) \rightarrow \mathbf{a}_{i\pm 1, i}$.

7 The force in full

The force on backbone vertex $\mathbf{a}_\alpha^P$ is the exact negative gradient of the master energy (5.11),

$$\mathbf{F}_\alpha^P = -\frac{\partial E}{\partial \mathbf{a}_\alpha^P}, \quad (7.1)$$

and because E is a finite sum of band contributions, so is the force: every band that geometrically involves $\mathbf{a}_\alpha^P$ donates a direct term and a boundary term (Lemma 5), each pushed back to the vertex through the same geometry that built the band. We assemble it in layers mirroring §4’s energy: differentials of the Green blocks, the band-endpoint Jacobians, the crossing-parameter sensitivities, and the adjoint geometry operator. Throughout, $\partial/\partial \mathbf{X}$ denotes the row gradient and contractions follow the summation convention.

7.1 Layer 0: differentials of the Green blocks

Differentiating the two building blocks (5.4) is immediate:

$$\boxed{\begin{aligned} \frac{\partial \mathcal{S}}{\partial X^m} &= \frac{1}{2} \epsilon_{mj} Y^j, & \frac{\partial \mathcal{S}}{\partial Y^m} &= \frac{1}{2} \epsilon_{im} X^i = -\frac{1}{2} \epsilon_{mi} X^i, \\ \mathcal{Q}'_\delta(\Delta\psi) &= \frac{1}{2} \delta^2(1 - \cos \Delta\psi), & \frac{\partial \mathcal{Q}_\delta}{\partial \delta} &= \delta(\Delta\psi - \sin \Delta\psi). \end{aligned}} \quad (7.2)$$

These are the only “physics” derivatives; everything below is geometry feeding $\mathbf{X}, \mathbf{Y}, \Delta\psi, \delta$.

7.2 Layer 1: band-endpoint Jacobians (direct vs. boundary)

A band endpoint is $\mathbf{X}_r = F(t_r)$ with F a feature parametrisation (5.2)–(5.3) and t_r either a fixed feature endpoint (0 or 1, hence a tangent point) or a crossing parameter. Splitting per Lemma 5:

Straight feature $E_i^P(s) = (1-s)\mathbf{a}_i^{P,+} + s\mathbf{a}_{i+1}^{P,-}$, with $\mathbf{e}_F := \mathbf{a}_{i+1}^{P,-} - \mathbf{a}_i^{P,+} = \dot{E}_i^P$:

$$\underbrace{\frac{\partial \mathbf{X}_r}{\partial \mathbf{a}_i^{P,+}} = (1-s_r)\mathbb{I}, \quad \frac{\partial \mathbf{X}_r}{\partial \mathbf{a}_{i+1}^{P,-}} = s_r\mathbb{I}}_{\text{direct (limits fixed)}} \quad \underbrace{\frac{\partial \mathbf{X}_r}{\partial s_r} = \mathbf{e}_F}_{\text{boundary (limit moves)}}. \quad (7.3)$$

Arc feature $K_i^P(\tau) = \mathbf{z}_i^P + \delta_P(\cos \varphi, \sin \varphi)$, $\varphi_r = \varphi_{0,i}^P + \tau_r \Delta\phi_i^P$, with the unit tangent $\boldsymbol{\rho}_r := (-\sin \varphi_r, \cos \varphi_r)$:

$$\underbrace{\frac{\partial \mathbf{X}_r}{\partial \mathbf{z}_i^P} = \mathbb{I}, \quad \frac{\partial \mathbf{X}_r}{\partial \varphi_{0,i}^P} = \delta_P \boldsymbol{\rho}_r, \quad \frac{\partial \mathbf{X}_r}{\partial \Delta\phi_i^P} = \tau_r \delta_P \boldsymbol{\rho}_r, \quad \frac{\partial \mathbf{X}_r}{\partial \delta_P} = (\cos \varphi_r, \sin \varphi_r)}_{\text{direct}} \quad \underbrace{\frac{\partial \mathbf{X}_r}{\partial \tau_r} = \Delta\phi_i^P \delta_P \boldsymbol{\rho}_r}_{\text{boundary}}. \quad (7.4)$$

The cap argument carries its own dependence $\Delta\psi = (\tau_1 - \tau_0)\Delta\phi_i^P$, so $\partial\Delta\psi/\partial\tau_1 = \Delta\phi_i^P = -\partial\Delta\psi/\partial\tau_0$ and $\partial\Delta\psi/\partial\Delta\phi_i^P = \tau_1 - \tau_0$.

7.3 Layer 2: crossing-parameter sensitivity operator

Every interior limit t_* is fixed by a scalar contact equation $\Phi(t_*; \text{both polygons}) = 0$, so by the implicit rule (6.2) $\partial t_*/\partial \theta = -(\partial_\theta \Phi)/(\partial_{t_*} \Phi)$. Collecting the three contact types from §5.7:

edge-edge: $\Phi = (\gamma(t_*) - \mathbf{R}) \times \mathbf{e}_B,$	$\frac{\partial \Phi}{\partial t_*} = \mathbf{e}_A \times \mathbf{e}_B,$
arc-edge: $\Phi = \mathbf{e} ^2 t_*^2 + 2(\mathbf{f} \cdot \mathbf{e}) t_* + \mathbf{f} ^2 - \delta^2, \quad \mathbf{f} = \mathbf{p} - \mathbf{z},$	$\frac{\partial \Phi}{\partial t_*} = 2 \mathbf{e} ^2 t_* + 2(\mathbf{f} \cdot \mathbf{e}),$
arc-arc: $\Phi = \gamma(\tau_*) - \mathbf{z}_B ^2 - \delta_B^2,$	$\frac{\partial \Phi}{\partial \tau_*} = 2 \delta_A \Delta \phi_A (\gamma(\tau_*) - \mathbf{z}_B) \cdot \boldsymbol{\rho}_*,$

(7.5)

and the numerators $\partial_\theta \Phi$ are read off the same expressions with θ a geometry datum $(\mathbf{e}_A, \mathbf{e}_B, \mathbf{z}, \delta, \mathbf{R}, \dots)$, each of which is itself a layer-3 function of the backbone vertices. Define the resulting map $\mathcal{D} : \theta \mapsto \partial t_*/\partial \theta$.

7.4 Layer 3: the adjoint geometry operator $\mathcal{G}_P^*$

This is `arcGeomGrad`: it receives the upstream sensitivities collected at the arc data of vertex i ,

$$\left(\frac{\partial E}{\partial \mathbf{a}_i^{P,-}}, \frac{\partial E}{\partial \mathbf{a}_i^{P,+}}, \frac{\partial E}{\partial \mathbf{z}_i^P}, \frac{\partial E}{\partial \varphi_{0,i}^P}, \frac{\partial E}{\partial \Delta \phi_i^P} \right),$$

and returns the three vertex forces $(\mathbf{f}_{i-1}, \mathbf{f}_i, \mathbf{f}_{i+1})$ using exactly the Jacobian blocks derived in §6.5. Consolidated, with $\hat{\mathbf{u}}, \hat{\mathbf{v}}, c, d, \ell$ as in (5.1) and $P[\hat{\mathbf{w}}]^{jk} = \delta^{jk} - \hat{w}^j \hat{w}^k$:

$\frac{\partial \ell}{\partial \hat{u}^i} = \frac{1}{1+d} [\delta_P \epsilon_{iq} \hat{v}^q - \ell \hat{v}_i],$	$\frac{\partial \ell}{\partial \hat{v}^j} = \frac{1}{1+d} [-\delta_P \epsilon_{jp} \hat{u}^p - \ell \hat{u}_j],$	(7.6)
$\frac{\partial \Delta \phi}{\partial \hat{u}^i} = d \epsilon_{iq} \hat{v}^q - c \hat{v}_i,$	$\frac{\partial \Delta \phi}{\partial \hat{v}^j} = -d \epsilon_{jp} \hat{u}^p - c \hat{u}_j,$	
$\frac{\partial a^{-m}}{\partial \hat{u}^i} = -\hat{u}^m \frac{\partial \ell}{\partial \hat{u}^i} - \ell \delta_i^m,$	$\frac{\partial a^{+m}}{\partial \hat{v}^j} = \hat{v}^m \frac{\partial \ell}{\partial \hat{v}^j} + \ell \delta_j^m,$	
$\frac{\partial z^m}{\partial \hat{u}^i} = -\hat{u}^m \frac{\partial \ell}{\partial \hat{u}^i} - \ell \delta_i^m - \delta_P \epsilon^m{}_i,$	$\frac{\partial z^m}{\partial \hat{v}^j} = -\hat{u}^m \frac{\partial \ell}{\partial \hat{v}^j},$	
$\frac{\partial \hat{u}^j}{\partial a_i^k} = \frac{1}{ \mathbf{u} } P[\hat{\mathbf{u}}]^{jk}, \quad \frac{\partial \hat{u}^j}{\partial a_{i-1}^k} = -\frac{1}{ \mathbf{u} } P[\hat{\mathbf{u}}]^{jk},$	$\frac{\partial \hat{v}^j}{\partial a_{i+1}^k} = \frac{1}{ \mathbf{v} } P[\hat{\mathbf{v}}]^{jk}, \quad \frac{\partial \hat{v}^j}{\partial a_i^k} = -\frac{1}{ \mathbf{v} } P[\hat{\mathbf{v}}]^{jk},$	

together with $\partial \mathbf{a}^-/\partial \hat{\mathbf{v}} = -\hat{\mathbf{u}} \otimes \partial \ell/\partial \hat{\mathbf{v}}$, $\partial \mathbf{a}^+/\partial \hat{\mathbf{u}} = \hat{\mathbf{v}} \otimes \partial \ell/\partial \hat{\mathbf{u}}$, the explicit dependences $\partial \mathbf{a}^\pm/\partial \mathbf{a}_i = \partial \mathbf{z}/\partial \mathbf{a}_i = \mathbb{K}$, and the start angle. Because $\mathbf{a}_i^{P,-} - \mathbf{z}_i^P = -\delta_P \hat{\mathbf{n}}$ exactly, the start angle collapses to a function of $\hat{\mathbf{u}}$ alone,

$$\varphi_{0,i}^P = \text{atan2}(-\hat{n}^2, -\hat{n}^1) = \text{atan2}(-\hat{u}^1, \hat{u}^2), \quad \frac{\partial \varphi_{0,i}^P}{\partial \hat{u}^i} \text{ via Lemma 3,} \quad (7.7)$$

a useful simplification: φ_0 does not depend on ℓ or δ_P . Folding (7.6) through the chain rule and collecting the projector-perpendicular gradients $\partial E/\partial \mathbf{u} = P[\hat{\mathbf{u}}] \cdot (\partial E/\partial \hat{\mathbf{u}})/|\mathbf{u}|$ yields the compact scatter the kernel emits,

$\mathbf{f}_{i-1} = -\frac{P[\hat{\mathbf{u}}]}{ \mathbf{u} } \frac{\partial E}{\partial \hat{\mathbf{u}}}, \quad \mathbf{f}_{i+1} = \frac{P[\hat{\mathbf{v}}]}{ \mathbf{v} } \frac{\partial E}{\partial \hat{\mathbf{v}}}, \quad \mathbf{f}_i = \mathbf{f}^{a-} + \mathbf{f}^{a+} + \frac{P[\hat{\mathbf{u}}]}{ \mathbf{u} } \frac{\partial E}{\partial \hat{\mathbf{u}}} - \frac{P[\hat{\mathbf{v}}]}{ \mathbf{v} } \frac{\partial E}{\partial \hat{\mathbf{v}}},$	(7.8)
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with $\mathbf{f}^{a^\pm}$ the explicit- $\mathbf{a}_i$ tangent-point parts. Write this map as $\mathcal{G}_P^*(i)$; it is the transpose of the geometry operator (5.1) and deposits onto the stencil $\{i-1, i, i+1\}$ for an arc, extended to $\{i-1, i, i+1, i+2\}$ for the flat section E_i^P (whose endpoints are the tangent points of vertices i and $i+1$).

7.5 The master force equation

For a single inside band, abbreviate the per-band gradient with respect to a generic geometry datum θ as the sum of a direct and a boundary part (Lemma 5, Layers 0–1):

$$\nabla_\theta^{\text{band}} := \underbrace{\frac{\partial \mathcal{S}}{\partial \mathbf{X}_0} \frac{\partial \mathbf{X}_0}{\partial \theta} \Big|_{\text{dir}} + \frac{\partial \mathcal{S}}{\partial \mathbf{X}_1} \frac{\partial \mathbf{X}_1}{\partial \theta} \Big|_{\text{dir}} + [\mathcal{Q}_{\delta_P}]_\theta}_{\text{direct}} + \underbrace{\sum_{r \in \{0,1\}} \left(\frac{\partial \mathcal{S}}{\partial \mathbf{X}_r} + [\mathcal{Q}]_{\mathbf{X}_r} \right) \frac{\partial \mathbf{X}_r}{\partial t_r} \mathcal{D}(t_r; \theta)}_{\text{boundary}}, \quad (7.9)$$

where $[\mathcal{Q}_{\delta_P}]_\theta$ is the cap's contribution through $\Delta\psi$ and δ_P (nonzero only for arc features), and $\mathcal{D}$ is the Layer-2 operator (7.5). The total force on vertex α of polygon P is then the negative sum of every band's ∇^{band} , routed to backbone vertices through the adjoint operator $\mathcal{G}_P^*$ and accumulated over the same pair/direction/ feature/band index set as the energy:

$$\mathbf{F}_\alpha^P = - \sum_{\{A,B\} \ni P} \sum_{(P',Q') \in \{(A,B), (B,A)\}} \sum_{i=0}^{N_{P'}-1} \left\{ \left[\mathcal{G}_{P'}^*(i) \circ \sum_{[\tau_0, \tau_1] \in \mathcal{I}(K_i^{P'}; \tilde{Q}')} \nabla^{\text{band}} \right]_\alpha + \left[\mathcal{G}_{P'}^*(i) \circ \sum_{[s_0, s_1] \in \mathcal{I}(E_i^{P'}; \tilde{Q}')} \nabla^{\text{band}} \right]_\alpha \right\} \quad (7.10)$$

The outer sum runs over the pair containing P (so P appears as a feature owner) *and* over the reversed direction (so P appears as the clipping body Q'); in the latter the only nonzero pieces are boundary terms, since $\Pi_{\tilde{P}}$ and the contact equations (7.5) carry P 's geometry — this is precisely how the reaction force on the other polygon's vertices arises, with $\sum \mathbf{F} = 0$ guaranteed by $\partial_\theta \Phi$ being shared between the two bodies up to sign. $[\cdot]_\alpha$ extracts the component of the $\mathcal{G}^*$ stencil that lands on vertex α .

How to read (7.10)

It is (5.11) differentiated, term by term, with the *same* index set. Each band emits ∇^{band} (7.9) — a Layer-0 Green differential times a Layer-1 endpoint Jacobian, plus a boundary piece carrying a Layer-2 crossing sensitivity — and $\mathcal{G}_{P'}^*$ (Layer 3, (7.8)) scatters it from arc data onto the backbone stencil. No band, no body, and no crossing is treated specially: edges drop the cap, the clipping-body direction keeps only boundary terms, and the projector $P[\hat{w}]$ in (7.8) makes every force automatically tangent to the constraint surface before SHAKE even runs.

7.6 Fully unrolled arc-band force

Take one fully interior arc band ($\tau_0 = 0, \tau_1 = 1$, so the limits are the tangent points and the boundary term vanishes). With $\varphi_r = \varphi_{0,i}^P + \tau_r \Delta\phi_i^P$, $\boldsymbol{\rho}_r = (-\sin \varphi_r, \cos \varphi_r)$, the band gradient with respect to the arc datum $\Delta\phi_i^P$ is, substituting (7.2) and Layer 1,

$$\nabla_{\Delta\phi_i^P}^{\text{band}} = \frac{1}{2} \epsilon_{mj} \left[(\tau_0 \delta_P \rho_0^m) X_1^j + X_0^m (\tau_1 \delta_P \rho_1^j) \right] + \frac{1}{2} \delta_P^2 (1 - \cos \Delta\psi) (\tau_1 - \tau_0), \quad (7.11)$$

and likewise $\nabla_{\mathbf{z}_i^P}^{\text{band}} = \frac{1}{2}\epsilon_{mj}(\delta_n^m X_1^j + X_0^m \delta_n^j)$ a 2-vector in n . Feeding $(\partial E/\partial \mathbf{z}_i^P, \partial E/\partial \varphi_{0,i}^P, \partial E/\partial \Delta \phi_i^P)$ and the tangent-point sensitivities into $\mathcal{G}_P^*(i)$ via (7.6)–(7.8), and recalling $\mathbf{z}_i^P = \mathbf{a}_i^P - \ell \hat{\mathbf{u}} + \delta_P \hat{\mathbf{u}}^\perp$, $\ell = \delta_P \tan(\phi/2)$, $\phi = \text{atan2}(\epsilon_{ab} \hat{u}^a \hat{v}^b, \hat{u}^a \hat{v}^a)$ with $\hat{\mathbf{u}}, \hat{\mathbf{v}}$ the normalised minimum-image vertex differences (5.1), expresses $\mathbf{f}_{i-1}, \mathbf{f}_i, \mathbf{f}_{i+1}$ purely in the backbone coordinates — the “gory” force, bottoming out at the same vertices as (5.11).

8 SHAKE: constraint projection

8.1 Constraints

A polygon with n vertices $\{\mathbf{x}_0, \dots, \mathbf{x}_{n-1}\}$ carries $n+1$ holonomic constraints — n equilateral edge lengths and one area:

$$C_k(\mathbf{x}) = |\mathbf{x}_{k+1} - \mathbf{x}_k| - L_k = 0 \quad (k = 0, \dots, n-1), \quad C_n(\mathbf{x}) = A(\mathbf{x}) - A_0 = 0. \quad (8.1)$$

Stack them as $\mathbf{C} \in \mathbb{R}^{n+1}$ with Jacobian $J \in \mathbb{R}^{(n+1) \times 2n}$, $J_{(k)} = \nabla_{\mathbf{x}} C_k$.

8.2 The simultaneous Newton step

We seek a correction $\Delta \mathbf{x}$ that drives all constraints to zero at once. Linearise: $\mathbf{C}(\mathbf{x} + \Delta \mathbf{x}) \approx \mathbf{C}(\mathbf{x}) + J \Delta \mathbf{x}$. Restrict the correction to the constraint subspace, $\Delta \mathbf{x} = -J^\top \boldsymbol{\lambda}$ (so it does no work against unconstrained directions), and demand $\mathbf{C}(\mathbf{x}) + J \Delta \mathbf{x} = 0$:

$$\mathbf{C}(\mathbf{x}) - J J^\top \boldsymbol{\lambda} = \mathbf{0} \quad \implies \quad \boxed{(J J^\top) \boldsymbol{\lambda} = \mathbf{C}(\mathbf{x}), \quad \Delta \mathbf{x} = -J^\top \boldsymbol{\lambda}.} \quad (8.2)$$

This is one Newton iteration on the constraint residual; iterate to convergence.

8.3 Jacobian entries

Edge constraint. With $\hat{\mathbf{e}}_k = (\mathbf{x}_{k+1} - \mathbf{x}_k)/|\mathbf{x}_{k+1} - \mathbf{x}_k|$, and using $\partial|\mathbf{w}|/\partial \mathbf{w} = \hat{\mathbf{w}}$ (Lemma 2),

$$\nabla_{\mathbf{x}_k} C_k = -\hat{\mathbf{e}}_k, \quad \nabla_{\mathbf{x}_{k+1}} C_k = +\hat{\mathbf{e}}_k, \quad \nabla_{\mathbf{x}_j} C_k = \mathbf{0} \quad (j \neq k, k+1). \quad (8.3)$$

Area constraint. From the shoelace gradient (2.5), $\nabla_{\mathbf{x}_k} C_n = \mathbf{g}_k = \frac{1}{2}(\mathbf{x}_{k-1} - \mathbf{x}_{k+1})^\perp$ (the ϵ -form (2.5) is the primitive definition; this $\perp$ -form follows from $\mathbf{u}^\perp = (-u^2, u^1)$).

8.4 The Gram matrix $M = J J^\top$

Entry $M_{kl} = \nabla C_k \cdot \nabla C_l$. We enumerate every case.

Edge diagonal. ∇C_k has $-\hat{\mathbf{e}}_k$ at $\mathbf{x}_k$ and $+\hat{\mathbf{e}}_k$ at $\mathbf{x}_{k+1}$, so $M_{kk} = |-\hat{\mathbf{e}}_k|^2 + |+\hat{\mathbf{e}}_k|^2 = 1 + 1 = 2$.

Adjacent edges ($l = k \pm 1 \pmod n$). C_k and C_{k+1} share only $\mathbf{x}_{k+1}$, where $\nabla_{\mathbf{x}_{k+1}} C_k = +\hat{\mathbf{e}}_k$ and $\nabla_{\mathbf{x}_{k+1}} C_{k+1} = -\hat{\mathbf{e}}_{k+1}$, so $M_{k,k+1} = \hat{\mathbf{e}}_k \cdot (-\hat{\mathbf{e}}_{k+1}) = -\hat{\mathbf{e}}_k \cdot \hat{\mathbf{e}}_{k+1}$.

Non-adjacent edges. No shared vertex $\Rightarrow M_{kl} = 0$ for $|k - l| > 1 \pmod n$, excluding the cyclic neighbours).

Area row/column. C_k touches $\mathbf{x}_k$ (with $-\hat{\mathbf{e}}_k$) and $\mathbf{x}_{k+1}$ (with $+\hat{\mathbf{e}}_k$); C_n has $\mathbf{g}_j$ at every $\mathbf{x}_j$. Hence

$$M_{n,k} = M_{k,n} = (-\hat{\mathbf{e}}_k) \cdot \mathbf{g}_k + (+\hat{\mathbf{e}}_k) \cdot \mathbf{g}_{k+1} = \hat{\mathbf{e}}_k \cdot (\mathbf{g}_{k+1} - \mathbf{g}_k).$$

Area diagonal. $M_{n,n} = |\nabla A|^2 = \sum_k |\mathbf{g}_k|^2$.

8.5 Solve and update

The $(n+1) \times (n+1)$ system $M\boldsymbol{\lambda} = \mathbf{C}$ is solved by unpivoted LU factorisation — M is symmetric and, for near-equilateral polygons, diagonally dominant in the edge block (diagonal 2, off-diagonals $-\hat{\mathbf{e}}_k \cdot \hat{\mathbf{e}}_{k+1} \in (-1, 1)$) so pivoting is unnecessary. The per-vertex displacement gathers the two edge multipliers adjacent to vertex k and the single area multiplier,

$$\Delta \mathbf{x}_k = -(\mathbf{J}^\top \boldsymbol{\lambda})_k = \lambda_k \hat{\mathbf{e}}_k - \lambda_{k-1} \hat{\mathbf{e}}_{k-1} - \lambda_n \mathbf{g}_k, \quad (8.4)$$

because $\mathbf{x}_k$ appears in C_k (coefficient $-\hat{\mathbf{e}}_k$, contributing $+\lambda_k \hat{\mathbf{e}}_k$ after the global minus), in C_{k-1} (coefficient $+\hat{\mathbf{e}}_{k-1}$, contributing $-\lambda_{k-1} \hat{\mathbf{e}}_{k-1}$), and in C_n (coefficient $\mathbf{g}_k$). Thread k updates vertex k ; all n run in parallel inside one block. Convergence is declared when $\max_k |C_k(\mathbf{x})| < \text{tol} = 10^{-15}$. If `maxIter` is exhausted the iteration count is multiplied by 10 and retried; a second failure rejects the FIRE step and reverts positions.

9 Force projection onto the constraint manifold

After the overlap force $\mathbf{F}$ is assembled it must be stripped of any component that would violate the constraints, so FIRE moves only along the manifold.

9.1 The constraint projector

The constraint subspace at $\mathbf{x}$ is the row space of J (the span of the constraint gradients). The orthogonal projector onto its complement is

$$\mathbf{F}_\perp = \left(\mathbb{I} - J^\top (JJ^\top)^{-1} J \right) \mathbf{F} = \mathbf{F} - J^\top (JJ^\top)^{-1} J \mathbf{F}. \quad (9.1)$$

Indeed $\Pi := J^\top (JJ^\top)^{-1} J$ satisfies $\Pi^2 = \Pi$, $\Pi^\top = \Pi$, and $\Pi J^\top = J^\top$, so $\mathbb{I} - \Pi$ is the orthoprojector onto $\ker J = (\text{row } J)^\perp$.

9.2 Edge block via the SVD of $G^\top$

Collect the edge-constraint gradients as the rows of $G \in \mathbb{R}^{n \times 2n}$,

$$G_{k,2k} = -\hat{e}_k^x, \quad G_{k,2k+1} = -\hat{e}_k^y, \quad G_{k,2(k+1)} = +\hat{e}_k^x, \quad G_{k,2(k+1)+1} = +\hat{e}_k^y, \quad (9.2)$$

all other entries zero. The constraint subspace they span lives in $\mathbb{R}^{2n}$; an orthonormal basis for it is the set of left singular vectors of $G^\top \in \mathbb{R}^{2n \times n}$. Compute

$$G^\top = U \Sigma V^\top, \quad U \in \mathbb{R}^{2n \times n}, \quad V \in \mathbb{R}^{n \times n}, \quad (9.3)$$

so that the columns $\mathbf{u}_1, \dots, \mathbf{u}_n$ of U are orthonormal and span $\text{range}(G^\top) = \text{row}(G)$.

Lemma 6 (Projector identity). If G has full row rank n then $G^\top (GG^\top)^{-1} G = UU^\top$.

Proof. With the thin SVD $G^\top = U \Sigma V^\top$ ($\Sigma \in \mathbb{R}^{n \times n}$ invertible for full rank), $G = V \Sigma U^\top$ and $GG^\top = V \Sigma^2 V^\top$, so $(GG^\top)^{-1} = V \Sigma^{-2} V^\top$. Then

$$G^\top (GG^\top)^{-1} G = (U \Sigma V^\top) (V \Sigma^{-2} V^\top) (V \Sigma U^\top) = U \Sigma \Sigma^{-2} \Sigma U^\top = UU^\top,$$

using $V^\top V = \mathbb{I}_n$. $UU^\top$ is the orthoprojector onto $\text{range}(U) = \text{row}(G)$. $\square$

Hence the edge part of the projection (9.1) is

$$\mathbf{F}_{\perp, \text{edge}} = \mathbf{F} - UU^\top \mathbf{F} = \mathbf{F} - \sum_{j=1}^n (\mathbf{u}_j \cdot \mathbf{F}) \mathbf{u}_j. \quad (9.4)$$

9.3 Area block by Gram–Schmidt

The area gradient $\mathbf{g}_A = \nabla A$ is generally *not* in the edge span, so orthogonalise it against the $\mathbf{u}_j$ and normalise:

$$\tilde{\mathbf{g}}_A = \mathbf{g}_A - \sum_{j=1}^n (\mathbf{u}_j \cdot \mathbf{g}_A) \mathbf{u}_j, \quad \hat{\mathbf{q}}_A = \frac{\tilde{\mathbf{g}}_A}{|\tilde{\mathbf{g}}_A|}. \quad (9.5)$$

$\hat{\mathbf{q}}_A$ is the unit constraint direction for the area, orthogonal to every edge direction. The full projection removes both blocks:

$$\mathbf{F}_\perp = \mathbf{F} - \sum_{j=1}^n (\mathbf{u}_j \cdot \mathbf{F}) \mathbf{u}_j - (\hat{\mathbf{q}}_A \cdot \mathbf{F}) \hat{\mathbf{q}}_A. \quad (9.6)$$

$\mathbf{F}_\perp$ has zero overlap with any constraint gradient, so a step along it neither stretches an edge nor changes the area to first order.

10 FIRE relaxation

FIRE (Fast Inertial Relaxation Engine; Bitzek *et al.*, 2006) is a damped molecular-dynamics minimiser: integrate Newtonian dynamics with the projected force, and whenever the motion has an uphill component, kill the velocity.

10.1 Velocity Verlet

With unit mass and force $\mathbf{F}(\mathbf{x})$, expand the trajectory in Taylor series about t_n (write $\mathbf{a} = \mathbf{F}$):

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_n dt + \frac{1}{2} \mathbf{a}_n dt^2 + \frac{1}{6} \ddot{\mathbf{x}}_n dt^3 + O(dt^4). \quad (10.1)$$

Velocity Verlet realises this with one force evaluation per step by introducing the half-step velocity and updating velocity with the trapezoidal average of the accelerations:

$$\mathbf{v}_{n+1/2} = \mathbf{v}_n + \frac{1}{2} \mathbf{a}_n dt, \quad (10.2)$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathbf{v}_{n+1/2} dt = \mathbf{x}_n + \mathbf{v}_n dt + \frac{1}{2} \mathbf{a}_n dt^2, \quad (10.3)$$

$$\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2} + \frac{1}{2} \mathbf{a}_{n+1} dt = \mathbf{v}_n + \frac{1}{2} (\mathbf{a}_n + \mathbf{a}_{n+1}) dt. \quad (10.4)$$

The position update (10.3) contains the acceleration exactly once: the $\frac{1}{2} \mathbf{a}_n dt^2$ is supplied by the half-kick (10.2) folded into $\mathbf{v}_{n+1/2} dt$, and no extra acceleration term is added. The scheme is time-reversible, symplectic, with local error $O(dt^3)$ and global error $O(dt^2)$.

SHAKE-coupled sequence. Each step: (i) half-kick (10.2); (ii) drift (10.3), then fold positions into $[0, 1]^2$; (iii) SHAKE-project $\mathbf{x}_{n+1}$ onto the constraint manifold; (iv) because SHAKE may move positions appreciably, rederive the half-step velocity from the *realised* displacement,

$$\mathbf{v}_{n+1/2}^{\text{re}} = \text{mic}(\mathbf{x}_{n+1}^{\text{SHAKE}} - \mathbf{x}_n) / dt;$$

(v) recompute and project $\mathbf{F}_{n+1}$; (vi) second half-kick $\mathbf{v}_{n+1} = \mathbf{v}_{n+1/2}^{\text{re}} + \frac{1}{2} \mathbf{a}_{n+1} dt$.

10.2 Inertial steering

Define the instantaneous power $\Pi = \mathbf{F}_{n+1} \cdot \mathbf{v}_{n+1}$.

Downhill, $\Pi > 0$. Bend the velocity toward the force while preserving its magnitude in the mixed direction,

$$\mathbf{v} \leftarrow (1 - \alpha) \mathbf{v} + \alpha |\mathbf{v}| \hat{\mathbf{F}}, \quad \hat{\mathbf{F}} = \mathbf{F}/|\mathbf{F}|. \quad (10.5)$$

After $n_{\min}$ consecutive downhill steps, accelerate: $dt \leftarrow \min(dt f_{\text{inc}}, dt_{\max})$, $\alpha \leftarrow \alpha f_{\alpha}$.

Uphill, $\Pi \leq 0$. Freeze and cool: $\mathbf{v} \leftarrow \mathbf{0}$, $dt \leftarrow dt f_{\text{dec}}$, $\alpha \leftarrow \alpha_0$, and reset the downhill counter.

Initial step	dt_0	10^{-3}
Maximum step	$dt_{\max}$	5×10^{-2}
Initial mixing	α_0	0.1
Mixing decay	f_{α}	0.99
Step increase	f_{inc}	1.1
Step decrease	f_{dec}	0.5
Min downhill steps	$n_{\min}$	5

10.3 Step ceiling for dense packings

Require that the displacement in one step at the largest step size stays a small fraction ε of the smallest edge, $v_{\max} dt_{\max} \lesssim \varepsilon L_{\min}$. The velocity built up over a step scales as $v_{\max} \sim F_{\max} dt_{\max}$, so $F_{\max} dt_{\max}^2 \lesssim \varepsilon L_{\min}$ and

$$dt_{\max} \lesssim \sqrt{\frac{\varepsilon L_{\min}}{F_{\max}}}, \quad \varepsilon = 10^{-3}, \quad (10.6)$$

with $L_{\min}$ the smallest target edge and $F_{\max}$ from the initial configuration.

11 Random shape generation by SLSQP

11.1 Parameterisation

Generate a random equilateral closed N -gon of prescribed shape index κ and area. Use the $N - 1$ free turning angles $\boldsymbol{\phi} = (\phi_0, \dots, \phi_{N-2}) \in (0, \pi)^{N-1}$, accumulate headings, and march unit edges:

$$\theta_k = \sum_{j=0}^{k-1} \phi_j, \quad \hat{\mathbf{e}}_k = (\cos \theta_k, \sin \theta_k), \quad \mathbf{v}_0 = \mathbf{0}, \quad \mathbf{v}_{k+1} = \mathbf{v}_k + \hat{\mathbf{e}}_k. \quad (11.1)$$

All edge lengths are 1, so the perimeter is exactly $P = N$.

11.2 Closure

The polygon closes iff $\mathbf{v}_N = \mathbf{v}_0$, i.e.

$$\sum_{k=0}^{N-1} \hat{\mathbf{e}}_k = \mathbf{0} \iff \sum_k \cos \theta_k = 0, \quad \sum_k \sin \theta_k = 0 \quad (\text{two equalities}). \quad (11.2)$$

11.3 Shape index

With unit edges $P = N$ and $\kappa = P/\sqrt{A} = N/\sqrt{A}$. Demanding $\kappa = \kappa_0$ is the single equality

$$A(\boldsymbol{\phi}) = \left(\frac{N}{\kappa_0}\right)^2 := A_{\text{unit}}. \quad (11.3)$$

11.4 Turning budget

By Theorem 2 a simple closed CCW polygon has total turning $\sum_{k=0}^{N-1} \phi_k = 2\pi$. This is imposed as an equality (a soft band about 2π may be used instead); the per-angle bounds $\phi_k \in (10^{-4}, \pi - 10^{-4})$ prevent degenerate or reflex corners.

11.5 The program and its gradients

$$\begin{aligned} \min_{\phi} \quad & \frac{1}{2} |\phi - \phi_0|^2 && \text{(stay near a random reference } \phi_0) \\ \text{s.t.} \quad & \sum_k \cos \theta_k = 0, \quad \sum_k \sin \theta_k = 0, \quad A(\phi) = A_{\text{unit}}, \\ & \sum_k \phi_k = 2\pi, \quad \phi_k \in (10^{-4}, \pi - 10^{-4}). \end{aligned} \tag{11.4}$$

SLSQP needs the constraint gradients. Since $\partial \theta_k / \partial \phi_m = \mathbb{K}[m < k]$,

$$\frac{\partial}{\partial \phi_m} \sum_k \cos \theta_k = - \sum_{k>m} \sin \theta_k, \quad \frac{\partial}{\partial \phi_m} \sum_k \sin \theta_k = + \sum_{k>m} \cos \theta_k. \tag{11.5}$$

For the area, $A = \frac{1}{2} \sum_k \epsilon_{ij} v_k^i v_{k+1}^j$ with $\partial \hat{e}_k / \partial \phi_m = \hat{e}_k^\perp \mathbb{K}[m < k]$ and $\partial \mathbf{v}_k / \partial \phi_m = \sum_{l < k} \partial \hat{e}_l / \partial \phi_m$, giving a closed sum the optimiser evaluates directly; the turning-budget gradient is the all-ones vector. The reference ϕ_0 is drawn from a Dirichlet distribution scaled to $[0, 2\pi]$, biasing toward convex-ish shapes. On success the unit-area polygon is rescaled and recentred to the target area,

$$\mathbf{v}_k \leftarrow \sqrt{\frac{A_{\text{tgt}}}{A_{\text{unit}}}} (\mathbf{v}_k - \bar{\mathbf{v}}), \quad \bar{\mathbf{v}} = \frac{1}{N} \sum_k \mathbf{v}_k, \tag{11.6}$$

and a uniformly random rotation is applied to remove orientation bias.

12 Bidisperse packing set-up

12.1 Target areas from packing fraction

Use $N_L = N_S = N_{\text{poly}}/2$ large and small polygons of equal shape index in the unit box, with the large perimeter a factor ratio above the small. Equal shape index, $\kappa = P/\sqrt{A}$, forces

$$\frac{P_L}{\sqrt{A_L}} = \frac{P_S}{\sqrt{A_S}} \implies \frac{\sqrt{A_L}}{\sqrt{A_S}} = \frac{P_L}{P_S} = \text{ratio} \implies A_L = \text{ratio}^2 A_S. \tag{12.1}$$

The backbone packing fraction is the total backbone area in the unit cell,

$$\phi = N_L A_L + N_S A_S = A_S (N_L \text{ratio}^2 + N_S) \implies A_S = \frac{\phi}{N_L \text{ratio}^2 + N_S}, \quad A_L = \text{ratio}^2 A_S. \tag{12.2}$$

12.2 Edge length and rounding radius

From $P = NL = \kappa\sqrt{A}$ the equilateral target edge is $L = \kappa\sqrt{A}/N$, used by SHAKE. The rounding radius is tied to the smaller polygon,

$$\delta = 0.15 L_S = \frac{0.15 \kappa \sqrt{A_S}}{N}, \tag{12.3}$$

which sits well inside the validity bound $\delta < L/(2 \tan(\pi/N))$ for both sizes.

12.3 Two-polygon reference numbers

For the two-polygon test fix the large area to the per-polygon area of a 64-polygon packing at $\phi = 0.8$, and obtain the small one from the perimeter ratio:

$$A_L = \frac{0.8}{64} = 0.012500, \quad A_S = \frac{A_L}{\text{ratio}^2} = \frac{0.012500}{1.96} \approx 0.006378 \quad (\text{ratio} = 1.4). \quad (12.4)$$

13 Periodic boundary conditions

All positions live in $[0, 1)^2$. Separations are reduced with $\text{mic}(x) = x - \lfloor x + \frac{1}{2} \rfloor$ componentwise, applied to each Cartesian component of an edge vector $\mathbf{u} = \mathbf{a}_k - \mathbf{a}_{k-1}$. For the interaction between polygon A (whose vertex i is being processed) and polygon B , all of B is placed in one consistent image: anchor B 's first vertex at the minimum-image position relative to $\mathbf{a}_i$ and translate the rest rigidly,

$$\mathbf{b}_0^{\text{MIC}} = \mathbf{a}_i + \text{mic}(\mathbf{b}_0 - \mathbf{a}_i), \quad \mathbf{b}_j^{\text{MIC}} = \mathbf{b}_0^{\text{MIC}} + (\mathbf{b}_j - \mathbf{b}_0). \quad (13.1)$$

This guarantees no edge of B straddles the wrap seam, so all the planar geometry above applies without periodic artefacts.

14 End-to-end computational flow

1. **Initialise.** Generate N_{poly} equilateral polygons of target area and shape index by SLSQP (§11.5); place them on a jittered grid; build neighbour cells.
2. **Force/energy.** For each vertex i of each polygon A in parallel: compute the arc geometry at i and $i+1$ (§3.2, §3.3); gather polygons $B \neq A$ in the 3×3 cell block; for arc K_i and edge E_i find all crossings with ∂B and integrate the signed inside contributions (4.3), (4.11); accumulate forces on $\{i-1, i, i+1, i+2\}$ through the arc-geometry chain rule (§6.5) and onto B 's vertices.
3. **Project the force.** SVD of $G^\top$ plus Gram–Schmidt of the area gradient (§9.2).
4. **FIRE step.** Half-kick $\rightarrow$ drift $\rightarrow$ fold $\rightarrow$ SHAKE $\rightarrow$ velocity rederivation $\rightarrow$ force $\rightarrow$ second half-kick $\rightarrow$ steering (§10.1).
5. **Iterate** until $\|\mathbf{F}_\perp\|_\infty < F_{\text{tol}}$ or the step budget is exhausted.

The whole pipeline rests on the four reusable facts collected at the start: the two Levi-Civita contractions, the projector derivative of a unit vector, the planar-argument derivative, and Green's theorem. Every formula above is a contraction of δ_{ij} and ϵ_{ij} with one of those four.